

Site of Digestion of Starch in the Gastrointestinal Tract of Dairy Cows and Its Effect on Milk Yield and Composition

JAMES E. NOCEK¹

Research and Applied Technology Department
Agway Inc.
Syracuse, NY
and Cooperative Research Farms
Guelph, ON, Canada

SEERP TAMMINGA

Department of Animal Nutrition
Agricultural University
Wageningen, The Netherlands

ABSTRACT

Physical and chemical processing of feed ingredients and feeding management strategies are major instruments of manipulating amount and site of starch digestion in the gastrointestinal tract. Generally, as rumen escape of starch increases, postruminal starch digestion increases, and there does not appear to be a limitation to intestinal starch digestion. However, the efficiency with which postruminal starch is digested decreases, which represents a limitation that warrants investigation. Even though digestible dietary starch is presented to the intestine, there is no net glucose absorption at the portal vein, and plasma glucose levels remain relatively unaffected. This result may be associated with the large metabolic requirement for postruminally absorbed glucose, which is preferentially used for oxidative metabolism at the visceral tissue level. In addition, peripheral glucose concentration is highly regulated. A possible implication is that the exogenous glucose supply may spare endogenously synthesized glucose for gut metabolism, allowing more to be directed to the mammary gland. Amino acids also may be spared (less metabolism of dietary and tissue amino acids in the gut). Current produc-

tion studies yield no clear evidence as to the benefits of postruminal digestion of starch to enhance milk yield or to change its composition. However, studies suggest that starch digested postruminally is used more efficiently for milk synthesis than that digested in the rumen.

(Key words: starch, digestion, milk yield)

Abbreviation key: DEI = digestible energy intake, IDS = intestinal digestible starch, MDV = mesenteric-drained viscera, NSC = nonstructural carbohydrates, PDV = portal-drained viscera, RDS = rumen degradable starch.

INTRODUCTION

The nutritional implications of carbohydrates has only recently emerged as important in dairy cattle. This is not restricted to structural carbohydrates, such as those in NDF (80, 81), but also nonstructural carbohydrates (NSC), including starches, soluble sugars, and other reserve carbohydrates (17, 97). The reasons for this are several fold. Carbohydrates are the most important source of energy for rumen microbes, and their behavior in the rumen differs greatly among sources. These differences are not limited to structural carbohydrates versus NSC but to origin and ruminal behavior patterns (135). Unique behavioral patterns result in different proportions of starch escaping undegraded from the rumen (101, 119, 150), resulting in a variable supply of starch for enzymatic digestion in the small intestine. In addition, fermentation patterns between structural carbohydrates and NSC (88)

Received August 31, 1990.

Accepted February 11, 1991.

¹Reprint requests.

and among different starch sources (72, 135) result in different nutrient profiles supplied to the host animal.

Modern approaches to feed evaluation indicate that energy (digestible, metabolizable, or net energy) and protein are no longer a sufficient nutrient description for optimal feeding. To progress, new approaches to feed evaluation that consider characterization of ruminal and postruminal digestion of protein and carbohydrates are needed in ration formulation (5, 26, 71, 91, 152).

Information on the behavior of starch in the digestive tract of sheep and beef cattle is rather well documented (100, 101), but comparatively little is known about its behavior in dairy cows; even less has been published on the consequences of starch for milk production and composition (132).

The aim of this review is to examine gastrointestinal starch digestion as it relates to 1) factors affecting and manipulation of ruminal and postruminal starch digestion, 2) net glucose absorption and metabolism, 3) limitations to postruminal digestion, and 4) opportunity for balancing ruminal and postruminal starch digestion to maximize milk and component yields.

Description of Nonstructural Carbohydrates

Nonstructural carbohydrates often are water-soluble and composed of sugars, starches, and pectins. Although pectins often are associated with the cell wall, they are precipitated by sodium lauryl sulfate of neutral detergent solution (3). They are not covalently linked to the lignified portions of the cell wall and are almost completely digested (90 to 100%) in the rumen. Galactans are unique to legumes in replacing starches as the carbohydrate reserve, and fructosans are the storage material in temperate grasses (143, 144). β -Glucans are peculiar to the cell wall of grasses and to the bran of oats, barley, and rye (1).

Although not all NSC is starch, it represents 70 to 80% of most cereal grains. Starch is composed of two major molecules, amylose and amylopectin. Amylose is a linear polymer of α -1,4-D-glucose units. Amylopectin is a branched polymer with linear chains of α -1,4-D-glucose that have an α -1,6 branch point every 20 to 25 glucose residues (27, 119). Starch is arranged in a highly organized fashion;

the amylose and amylopectin molecules are held together by hydrogen bonding. Starch granules are insoluble in cold water and swell reversibly. Starch granules are "pseudocrystals" that contain regions of organized "crystalline" form (primarily amylopectin) and also nonorganized "amorphous" areas. The crystalline regions are quite resistant to water infiltration, whereas the water moves freely through the amorphous areas (27, 119).

Swelling is a process resulting when starch is exposed to water and gradually heated (55°C). The granules swell and take up water equal to 50% of their weight. Swelling is reversible after cooling and drying. However, as more heat is applied (60 to 80°C), starch undergoes irreversible swelling or gelatinization, in which the granules lose their crystallinity (27). This process is associated primarily with the amorphous regions; however, as heat and moisture are added, the crystalline areas also are affected.

Retrogradation of starch is the reassociation of starch molecules separated after gelatinization. Hydrogen bonding between amylose and amylopectin is reestablished, although retrograded starch does not regain the native starch character. Retrogradation can be reversed to some degree by heating (119).

The aforementioned processes are applied to many feedstuffs through manufacturing and fermentation. Reconstitution and subsequent ensiling of cereal grain is an example of these processes. The amount of water used in the process and the quality of fermentation (oxygen limitation, packing, etc.) can influence the degree of crystallinity in the final product. Gelatinizing is a process in which starch granules are ruptured completely by a combination of moisture, heat, pressure, and, in some cases, mechanical shear force. Expanding is a process that subjects the grain to moisture, pressure, and heat to gelatinize the starch. Extrusion, steamflaking, and rolling processes can result in gelatinized starch molecules. The type and degree of processing are critical in altering digestibility and use of nutrients by the animal.

Importance of Carbohydrates in Microbial Synthesis

Microorganisms depend on carbon skeletons and the availability of energy (ATP) for protein synthesis. Even though some ruminant

systems are based on fermentable or digestible OM (34, 56), rumen microorganisms are able to grow only on carbohydrate or secondary products of carbohydrate digestion. Rumen bacteria have specific requirements for maintenance before growth can occur. Bacterial yield depends on their specific growth rate and on the fractional degradation rate of carbohydrate. This relationship is represented in the following equation (52): $1/Y = M/k + 1/Y_g$, where Y = yield of bacteria (wt/wt of carbohydrate fermented); M = grams of carbohydrate per gram of bacteria per hour; k = growth rate per hour; Y_g = maximum yield (40 g of bacteria/100 g of OM fermented).

Organisms that degrade fatty acids grow too slowly to sustain a viable niche in the rumen, and only the glycerol portion of the fat is fermented (50). Although amino acids are fermented, this is an uneconomical process. Microbial growth on amino acids alone often is not possible because ATP synthesis is slow when this substrate is utilized (15, 42, 43). Systems have been developed to characterize the quantity of digestible carbohydrate in the rumen to predict microbial yield. Workers in Denmark (50) have identified digestible carbohydrates as the sum of digestible N-free extract and digestible crude fiber. Researchers in France (51) utilize digestible OM minus protein escaping rumen degradation and total lipid. In the recently developed Dutch system, a correction is made for starch escaping rumen degradation. These systems were shown to be better predictors of microbial yield than simply fermentable OM (50).

Rumen Degradation of Nonstructural Carbohydrates

Great strides in increasing milk production have been achieved through formulating rations for specific amounts of NSC and structural carbohydrate (17, 70, 94). These responses were attributable not only to optimizing structural carbohydrate and NSC balance but to inherently synchronizing protein and carbohydrate availabilities in the rumen. In all instances, however, the manipulated carbohydrate component was a quantitative measure. Generally, NSC are rapidly and entirely degraded in the rumen. For many ingredients, this is a valid assumption, at least to the degree

that regulating carbohydrate concentration by quantitative means causes a physiological response that results in increased milk production.

Table 1 identifies various measures used to determine the amount of NSC (quantitative) for several feedstuffs. An approach is needed that considers the nature of NSC (qualitative) in feedstuffs (i.e., degradability of carbohydrate types in the rumen). This approach should yield a more precise estimate of ruminal and postruminal energy contribution from both NSC and structural carbohydrate sources, account for differences associated with ruminal behaviors, and allow synchronization with other nutrients (especially protein). As with any concept of this nature, a data base to characterize degradability characteristics of ingredient carbohydrates (i.e., rate and extent of digestion) is essential; an adequate one has not yet been assembled, although several attempts have been made (8, 37, 78). Development of a standard technique has been a critical missing link in qualitatively describing rumen degradable carbohydrate in that several dynamic components are involved (93).

Table 2 lists ruminal starch degradabilities determined by either in situ or in vivo methodologies for several ingredients. When evaluating degradabilities by different techniques, one must identify and evaluate the objective for the test procedure used. Several in vitro techniques utilize enzymes and short incubation times to establish rates of starch digestion (60%/min to 8%/h). These numbers are appropriate for comparative purposes within a study; however, they may not represent actual rates or be comparable for a given ingredient across studies and techniques.

Although starch is not the only NSC source in feed ingredients, its quantitative and qualitative determinations represent the largest data base. The relationship between starch content (percentage of DM) and rumen degradability of starch (percentage of total starch) determined by both in situ and in vivo methods was $\text{starch} = 132.9 - 1.09 [\text{rumen degradable starch (RDS)}]$, $r^2 = .29$, $P < .02$ (Figure 1). Although significant, this relationship suggests that concentration does not account for much of the variation associated with degradability. Although the number of determinations was few, the relationship between in situ (percentage of

TABLE 1. Various determinations of nonstructural carbohydrate for grain, by-products, and forage sources (% of DM).

Feedstuffs	Total nonstructural carbohydrate ¹				Nonstructural carbohydrate ²				Starch ³			
	n ⁴	$\bar{X}$	SD	Refer- ence	n	$\bar{X}$	SD	Refer- ence	n	$\bar{X}$	SD	Refer- ence
Grains and by-products												
Alfalfa, dehydrated	...				1	22.5	...	(89)	...			
Bakery waste	...				1	55.0	...	(89)	...			
Barley	3	69.4	5.7	(9, 14)	4	58.0	2.8	(9, 82, 89)	15	60.6	10.5	(9, 85, 136, 154)
Beans, horse	...				1	63.0	...	(89)	5	39.2	3.5	(136)
Beet pulp	3	10.7	.2	(69, 82, 128)	2	29.1	1.8	(82, 89)	1	1.5	...	(136)
Brewers dried grains	4	15.7	1.7	(14, 69, 82, 128)	2	13.5	.1	(82, 89)	1	3.8	...	(138)
Canola meal	...				1	14.1	...	(89)	2	13.5	3.5	(136)
Citrus pulp	2	27.2	.1	(69, 82)	2	60.9	.2	(82, 89)	1	1.4	...	(136)
Coconut meal	...				1	44.0	...	(89)	1	8.0	...	(136)
Corn												
Corn cobs	1	15.5	...	(130)	2	10.3	6.9	(89, 130)	...			
Grain, ground	6	74.6	1.7	(9, 14, 69, 82, 128, 130)	3	73.7	2.1	(9, 82, 89)	24	76.1	8.8	(31, 134, 136, 145, 154)
HM ⁵ shell	...				1	75.1	...	(89)	6	79.0	9.1	(31)
HM ground ear	...				2	58.4	1.4	(82, 89)	...			
Hominy	3	59.5	7.8	(14, 69, 128)	2	52.0	5.6	(82, 89)	1	31.0	...	(136)
Solubles	1	13.9	...	(69)	1	30.3	...	(89)	...			
Gluten feed	5	21.6	4.3	(69, 82, 128)	2	23.3	5.3	(82, 89)	1	40.3		(136)
Gluten meal	4	19.4	2.2	(14, 69, 82, 128)	2	14.5	.2	(82, 89)	1	20.5		(136)
Cottonseed												
Hulls	...				1	1.4	...	(89)	...			
Meal	...				2	14.7	2.0	(82, 89)	1	1.3		(136)
Whole	...				1	8.2	...	(89)	...			
Distillers grains												
Corn	4	10.5	1.7	(14, 69, 82, 128)	2	14.1	2.5	(82, 89)	...			
Milo	...				1	40.0	...	(89)	...			
Feathermeal	...				...				1	7.4	...	(136)
Fish meal	...				1	2.0	...	(89)	...			
Linseed meal	...				2	26.8	2.7	(82, 89)	2	2.8	8.5	(136)
Millet	...				1	63.3	...	(89)	2	63.7	2.9	(136)
Oats	3	56.9	.1	(69, 82, 128)	2	46.5	.8	(82, 89)	7	42.1	3.7	(85, 136)
Peanut meal	3	59.7	4.2	(69, 130)	2	25.2	1.6	(82, 89)	2	4.4	1.1	(136)
Potato meal	...				1	80.5	...	(89)	1	65.1	...	(136)
Rice												
Bran	...				1	25.0	...	(89)	1	26.9	...	(136)
Grain	...				1	73.8	...	(89)	1	84.8	...	(136)
Sunflower meal	...				2	26.6	...	(89)	2	13.0	6.0	(136)
Sorghum grain	...				2	69.9	5.5	(82, 89)	4	75.4	8.7	(4, 136)
Soya												
Beans, whole	...				2	28.4	6.3	(82, 89)	4	9.0	1.1	(136)
Hulls	3	21.2	.2	(69, 82, 128)	2	14.2	.2	(82, 89)	...			

(continued)

TABLE 1. (continued) Various determinations of nonstructural carbohydrate for grain, by-products, and forage sources (% of DM).

Feedstuffs	Total nonstructural carbohydrate ¹				Nonstructural carbohydrate ²				Starch ³			
	n ⁴	$\bar{X}$	SD	Refer- ence	n	$\bar{X}$	SD	Refer- ence	n	$\bar{X}$	SD	Refer- ence
Meal (49% CP)	3	9.3	.4	(14, 128)	2	27.3	.1	(82, 89)	...			
Meal (48% CP)	2	12.9	4.8	(69, 82)	2	25.2	3.1	(82, 89)	6	1.3	.6	(136)
Tapioca	...				...				2	74.5	2.7	(136)
Wheat Bran	3	28.0	9.2	(14, 69, 82)	2	20.8	.3	(82, 89)	2	13.3	.2	(136)
Flour	...				...				1	77.2	...	(136)
Grain	...				2	69.0	2.8	(82, 89)	6	64.1	3.7	(85)
Middlings	3	48.3	.2	(69, 82, 128)	2	35.0	.9	(82, 89)	1	23.8	...	(136)
Whey dried	1	89.2	...	(68)	1	75.3	...	(89)	...			
Forages												
Alfalfa Hay	3	8.7	.2	(8, 14)	6	24.3	1.1	(8, 9, 82)	1	2.2	...	(118)
Silage	5	5.8	3.8	(9, 14, 69)	2	24.7	.8	(89)	2	8.1	7.2	(118, 154)
Corn silage	3	32.0	3.8	(8, 9, 14)	6	36.1	7.0	(8, 9, 82, 89)	5	39.4	9.5	(85, 136)
Grass, mixed	2	5.1	2.3	(14, 82)	...				...			
Oat silage	...				1	14.9	...	(89)	2	9.5	6.8	(85, 118)
Timothy hay	1	1.4	...	(69)	5	10.1	2.8	(82, 89)	2	2.9	1.9	(85)
Wheat straw	...				1	1.8	...	(89)	...			

¹Literature value determined by enzymatic methods using the procedure of Smith (126) with Taka-diastase.

²Determined by difference equations: 100 - (NDF + Protein + Lipid + Ash), analytical values were obtained from NRC (89) and literature values.

³Literature values determined by using various enzymatic methods with bacterial amylases.

⁴There may be more than one observation per cited reference.

⁵High moisture.

total starch) and in vivo starch (percentage of starch intake) degradabilities was in situ = $-89.8 + 2.01$ (in vivo), $r^2 = .65$, $P < .005$ (Figure 2). This suggests that methodologies to determine RDS are relatively comparable but that more comparisons are needed.

Dynamic Factors Influencing Rumen Starch Degradability and Postruminal Carbohydrate Delivery

The ruminal digestion process is a dynamic sequence and synergy of events that influences fermentation end product and delivery of carbohydrate sources postruminally. The degree to which these processes are manifested in the system is highly dependent on the type and amount of carbohydrate sources consumed by

the animal. Rate and extent of digestion are critical components in the process. Other factors influencing digestion are grain processing, particle size reduction, and particle passage rate.

Grain Processing. The effect of processing on utilization of the starch of cereal grains has been reviewed extensively (35, 137). Generally, processing is associated with improvements in efficiency of nutrient utilization by ruminal microorganisms and the total tract. The term processing can be subdivided into physical and chemical methodologies. Physical processing methods usually consist of breaking, cracking, grinding, rolling, or pelleting dried grains. Physiochemical modifications involve the application of heat and water, which act to hydrate and swell amorphous and crystalline

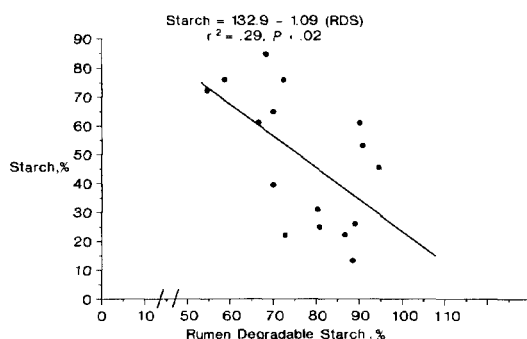


Figure 1. Relationship between starch concentration (quantitative, percentage of DM) and rumen degradable starch [(RDS) qualitative, percentage of total starch], in various feedstuffs.

structures of the starch granule. This alteration in structure enhances amylolytic digestion by both microorganisms and pancreatic enzyme action. The degree of moist heat application to grain, in addition to physically reducing particle size, has benefits greater than either process alone (137). Chemical treatments have been shown to enhance or retard ruminal degradation, depending on the chemical and its concentration used.

Decreasing corn flake density resulted in a linear decrease in ruminal pH and an increase in post-ruminal and total tract digestibilities (158). Galyean et al. (31) further indicated that processed corn (steamflaking and high moisture ensiling) resulted in increased starch digestibility beyond that of reducing particle size alone. However, grinding was more beneficial in enhancing DM digestion of unprocessed compared with processed corn. Nocek (92) showed that high moisture, shelled, and ear corn had a greater extent of ruminal digestion than the dry forms and that grinding generally increased the extent of degradation for all forms of corn. A similar effect was observed with pelleting (135).

The native protein configuration as it relates to starch granules can influence starch degradability. Starch granules can be embedded in a protein matrix of the endosperm of cereal grains, especially in corn and sorghum (119). Processing can disrupt this matrix and allow starch to be more accessible to enzymatic digestion. However, certain types of processing

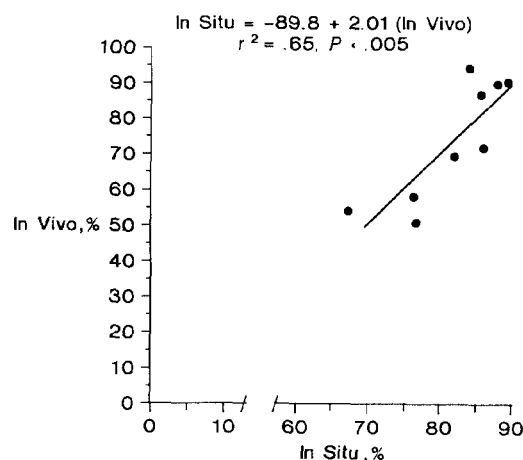


Figure 2. Relationship between in situ (percentage of total starch) and in vivo (percentage of starch intake) methods for determining rumen degradable starch for various feedstuffs.

(those causing gelatinization) may lead to formation of indigestible starch-protein complexes (138).

Most processing methods increase ruminal starch digestion, which usually increases percentage of starch digestion in the small intestine as well (101). Under some situations, it may be beneficial to increase the supply of starch to the lower gut. Although feeding of whole or coarsely cracked corn decreased rumen starch digestibility, compared with more finely ground corn, the larger corn particles may not be enzymatically digested in the small intestine. In addition, the passage of larger, more dense corn particles from the intestinal tract is accelerated, which decreases total tract digestion. Thus, the initial objective of shifting rather than limiting starch digestion may be nullified (101).

Several chemicals have been added to cereal grains to determine their influence on starch flow and digestibility in the digestive tract of ruminants. Fluharty and Loerch (25) reported that ruminal corn starch digestion in sheep decreased 30 and 41.5% with the addition of 1 and 2% formaldehyde, respectively, but total tract digestion was not affected. Glyoxal, masonex, propionaldehyde, and tannic acid were ineffective in reducing in vitro DM digestibility of corn. In another study, (98) treating corn with formaldehyde resulted in a 38% reduction in ruminal starch digestion and

TABLE 2. Rumen degradable starch (% of total starch) values for various feedstuffs determined by in vitro, in situ, or in vivo methods.

Feedstuffs	In situ and in vitro					In vivo				
	n ¹	$\bar{X}$	SD	Range	Reference	n	$\bar{X}$	SD	Range	Reference
Barley, ground	3	89.9	6.2	83.2-97.0	(36, 136)	4	87.9	4.6	82.4-93.0	(135, 147, 148, 149)
Rolled	...					8	87.2	2.5	73.2-90.3	(86, 124, 129, 133, 151)
Beans, faba	3	74.2	4.4	69.9-78.7	(135)	...				
Canola meal	1	72.5	...		(135)	...				
Corn										
Whole	...					5	62.6	9.9	58.9-75.0	(2, 30, 67, 101, 140)
Cracked	...					2	65.0	5.6	61.0-68.9	(101, 140)
Ensiled, shelled	1	72.0	...		(31)	3	86.0	3.0	83.0-89.0	(29, 101)
Ground	5	58.4	5.1	53.1-67.0	(31, 36, 136)	11	76.4	12.1	51.4-93.0	(30, 101, 135, 145)
Rolled, wet	...					1	68.3			(158)
Rolled	1	51.0	...		(31)	5	76.8	5.0	72.0-84.0	(2, 11, 29, 101, 129)
Steam-flaked	1	87.0	...		(31)	5	85.6	5.0	82.0-94.0	(2, 11, 29, 67, 101)
Hominy	2	66.2	6.4	61.6-70.7	(136)	...				
Silage	1	69.6	...		(136)	4	82.0	9.5	83.0-93.0	(101)
Corn gluten feed	3	80.6	.8	79.7-81.3	(101)	...				
Corn gluten meal	1	86.5	...		(101)	...				
Millet	1	73.7	...		(101)	...				
Oats	2	94.3	6.6	89.6-99.0	(36, 136)	2	84.0	11.1	76.1-91.8	(85, 124)
Potatoes meal	2	69.7	13.7	60.0-79.4	(136)	...				
Peas	2	75.8	2.1	74.3-77.3	(136)	...				
Sorghum										
Grain, ground	2	54.2	4.4	51.0-57.3	(136)	9	67.3	17.5	42.0-91.0	(39, 40, 41, 79)
Ensiled	...					1	86.2	...		(101)
Rolled	...					9	64.0	9.6	45.0-76.0	(32, 39, 40, 41, 57, 101, 129)
Steam-flaked	...					7	82.6	8.3	71.0-96.0	(32, 40, 44, 57, 79)
Rice										
Bran	3	71.3	3.3	67.5-73.5	(136)	...				
Grain	1	68.0	...		(136)	...				
Rye	...					1	90.2	...		(124)
Soybean meal (44% CP)	1	80.0	...		(136)	...				
Tapioca	3	82.7	5.3	76.7-86.8	(136)	...				
Wheat										
Bran	1	88.2	...		(136)	...				
Grain	3	90.5	7.3	86.1-99.0	(36, 136)	5	89.3	2.1	85.6-91.7	(85, 124)
Flour	2	88.3	.6	87.9-88.7	(136)	...				
Middlings	2	88.8	.8	88.2-89.4	(136)	...				

¹There may be more than one observation per cited reference.

increased starch digested in the small intestine; however, total tract digestibility was unaffected. Morgan et al. (86) treated barley with formaldehyde and indicated that starch flow to the abomasum was unaffected, but there was a trend towards higher protein flow on the treated diets.

Mandell et al. (73) treated whole barley with anhydrous ammonia (5% wt/wt) and compared this with rolled barley at 15 and 30% moisture. Ruminant OM digestion was lower for ammoniated whole compared with rolled barley (48.0 vs. 54.0% of OM intake). However, OM digestion in the small intestine of the

animals fed ammoniated whole barley was higher than for rolled barley (23.8 vs. 18.5% of OM intake) with no difference in total tract OM digestion. Robinson and Kennelly (116, 117) showed that total tract apparent digestibility of starch was not influenced by ammonia treatment of high moisture barley (0, .63, 1.3, and 1.95 g of ammonia/100 g). However, milk yield and protein and lactose concentrations increased linearly as ammonia level increased. Their findings suggest that shifting the site of starch digestion from the rumen to the small intestine resulted in a greater energetic efficiency of starch digestion and utilization. Although ammoniation did not influence DMI, cows tended to eat more rapidly as the level of ammonia treatment increased (117). Mandell et al. (73) indicated palatability problems with feeding ammoniated barley to beef cattle.

Alkaline (sodium hydroxide) treatment (4.5%) of oats was compared with whole and rolled oats in dairy cows (85). Cows fed treated oats tended to have higher DMI and produced more milk fat and protein. After 12 to 24 h of ruminal incubation, treated oats exhibited less DM disappearance than rolled oats with a more steady rate of *in situ* digestion. However, daily excretion of fecal starch from cows fed alkali-treated oats was nearly twice that from those fed whole and 10 times that from those fed rolled oat diets. Calculated total tract digestibility was highest for rolled and lowest for alkali-treated oats. Rumen ammonia N concentrations were highest for cows fed alkali-treated oats. The authors indicated that an N-energy interaction in the rumen may be associated with these occurrences.

Rate and extent of ruminal degradation of cereal grain can be either increased or decreased, depending on process utilized. Therefore, site of digestion depends on the type and degree of processing. The type of grain processed and the proportion of that grain in the diet also can influence digestion site. In addition, there may be differences in the manner in which cereal grains are utilized by body tissues because of unique composition characteristics once they are digested and absorbed.

Particle Size Reduction and Passage. Particle size and outflow from the rumen have a major influence on the digestive process. Breakdown of long feed particles has been

suggested as the rate-limiting step in ruminant digestion (87). Major modes of particle breakdown occur through mastication, rumination, microbial fermentation, and detrition by ruminal musculature activity (87, 105, 127). Particle size reduction effectively increases surface area and makes particles more fragile and accessible for digestion. Decreasing particle size also increases particle density and specific gravity (95). These processes cause stratification of particles within the rumen; small particles sink (higher density) and are passed out of the rumen, and larger particles (low density) rise within the ruminal strata and thus become further subjected to microbial breakdown and possibly rumination. It also has been suggested that chemical as well as physical composition will affect extent of digestion and particle breakdown (22).

Most research on particle size reduction thus far has concentrated on forages, which are high in structural carbohydrate, with relatively little attention to grains or grain by-products. However, because of the dynamics of rumen behavior, site and extent of starch digestion may be manipulated through the degree and extent of structural carbohydrate digestion.

Ewing et al. (24) evaluated particle breakdown and passage of corn. Their pool sizes were 1) >8; 2) <8 and >4.79; 3) <4.79 and >1.9; and 4) <1.9 mm. No particle reduction was detected for the largest pool size, whereas the second and third largest had rates of reduction of 1.9 and 1.8%/h, respectively. Particle reduction in the smallest pool was by fermentation and, hence, could pass out of the rumen as such. They indicated that decreasing corn particle size resulted in increased rates of passage from the rumen. Information with other NSC ingredient sources is not readily available. Particle breakdown is a dynamic process. Breakdown of large particles does not necessarily mean progression into the next smaller particle size pool. Particle size reduction is a function of initial particle size, rate of DM digestion, and quantity of DM that represents a given pool size (95, 96).

Dixon and Milligan (19) reported a progressive decline in fractional outflow from the rumen with increasing particle size and negligible outflow of particles >4 mm. They also suggested that, if ruminal contents were to be considered as two pools, a division at 3.2

mm would be appropriate. Poppi et al. (106) suggested that critical particle size for passage out of the rumen was 1.8 mm.

Initially, feed particles are less dense than water (specific gravity <1.0). As hydration takes place, density of feed particles increases and hydration potential decreases. In addition, physical compaction associated with chewing, digestion, etc. causes particle density to increase. Chemical composition influences hydration and, therefore, density. Cellulosic materials resist hydration, whereas carboxyl groups in hemicellulose and phenolic groups of lignin contribute to ion-exchange capacity and hydration (77). Heavy particles have been shown to shift through rumen stratification to a greater degree and to become more accessible for passage than lighter particles (20). No critical specific gravity has been characterized for passage of particles through the rumen. Campling and Freer (7) showed that passage of plastic particles was maximized at a specific gravity range of 1.1 to 1.2. Ehle (21) showed a considerable increase in rumen turnover rate of particles when bulk density was increased from 1.165 to 1.242 g/ml.

Feed particle size and reduction rate can influence density, which influences rate of passage out of the rumen. Rates of particulate and fluid passage will strongly influence whether or not starch will be digested in the rumen or escape to the lower gut.

Ruminal VFA Patterns. The pattern in which VFA (acetic, propionic, and butyric acids) are produced and present in the rumen is a reflection of substrate composition. Murphy et al. (88) showed important differences in VFA patterns produced from structural carbohydrates or NSC, which can be explained as reflecting differences in rate of carbohydrate degradation. Because of differences in rate of degradation among different starch sources, one also would expect differences in VFA patterns in the rumen of dairy cows after feeding different starch sources. Table 3 lists rumen VFA patterns for several feeding regimes. Although starch source influenced acetate and propionate concentration for some diets, differences were small.

Figure 3 summarizes the major components of ruminal carbohydrate digestion. The key to the entire process is the interdependence of these singular events, the fermentation end

products generated, and the passage of nutrients to the lower gut.

Postruminal Starch Delivery

The dynamics of postruminal starch digestion have received very little attention compared with the rumen. However, the same rules apply: 1) feed particles must pass through the tract; 2) feed particles must be reduced in size or digested; and 3) end products of digestion must be absorbed (Figure 4). Major differences from the rumen are that mammalian enzymatic digestion predominates, and substrate uptake for the host is primarily through active rather than passive transport. These events are dependent on the amount of starch and the rate at which it is delivered.

Nylon bag incubation studies (135, 136) reveal important differences among and within starch sources regarding rumen escape potential (Table 4). In these studies, it was assumed initially that starch being washed out of the nylon bags was degraded completely at an infinite rate. Comparing results of nylon bag studies with those of in vivo experiments for starch escaping the rumen (Table 2) showed an underestimation of the proportion of starch escaping from slowly degraded corn but an overestimation of rapidly degraded barley, oats, and wheat. The former has been observed earlier (23) and may be caused by the absence of particle size reduction because of chewing during eating or rumination, which, in the case of corn, could be a significant factor. The lower values for the other grains may result from the fact that a small proportion of starch washed out of the bags is not degraded in the rumen and is passed to the small intestine. In addition, some starch may leave the rumen incorporated in microbial cells (38). The net result of these two processes was somewhat arbitrarily set at 10% of the proportion that was washed out. Taking this into account, the escape values of starch for grains based on nylon bag incubations (Table 4) were similar to those obtained in vivo (Table 2, data for lactating dairy cows only). A further validation was made by comparing starch escape values for mixtures of starches of different origin (grain by-products, legume seeds, and tapioca); values were calculated on the basis of the data in Table 4 with the results of in vivo measure-

TABLE 3. Ratios of VFA in the rumen of dairy cows after feeding different starch sources.¹

Roughage	Concentrates	Acetate	(mol/100 mol)		Total VFA (mmol/L)	Reference
			Propionate	Butyrate		
Alfalfa hay	Barley	58.0	26.0 ^a	12.4	...	(6)
	Milo	62.6	21.4 ^b	12.2		
Grass hay	Barley ²	61.2 ^a	21.9 ^a	11.0	...	(133)
	Corn	62.9 ^a	21.0 ^a	11.6		
	Barley ³	43.3 ^c	41.0 ^b	9.2		
	Corn	53.0 ^b	28.0 ^a	11.2		
Alfalfa hay	Barley	64.6	20.4	11.8	127.6	(16)
	Corn	64.0	21.9	12.0	118.6	
Alfalfa silage	Barley	58.2	24.5 ^a	13.4	...	(154)
	Corn	61.1	20.8 ^b	13.7	...	
Alfalfa hay plus corn silage	Barley	58.2	28.2	9.9 ^b	78.9	(8)
	Corn	58.2	26.6	12.3 ^a	78.7	
Alfalfa-grass silage plus corn silage	Barley	57.3 ^b	31.0	9.2	120.5	(78)
	Corn	60.0 ^a	26.4	10.9	111.8	
Grass hay	Corn	63.0	18.4	15.2	108.1 ^a	(135)
	Barley	61.5	18.4	15.0	106.1 ^b	
	Tapioca	62.2	18.5	14.7	108.9 ^a	
Grass silage plus corn silage	Corn	59.9	24.6	12.7	142.0	(18)
	Barley	60.6	24.6	12.0	142.0	

^{a,b,c}Within each study, means in the same column which do not have the same superscript differ ($P < .05$).

¹Within each study, the same forage sources were fed among grain source treatments.

²Diets contained 60% barley or corn and 40% grass hay.

³Diets contained 90% barley or corn and 10% grass hay.

ments in dairy cows equipped with reentrant cannula (Tammenga, unpublished results). In this comparison, it was assumed that microbial OM contained 20% carbohydrates, of which 25% was starch. A further assumption was that pelleting reduced the escape of starch to the small intestine by 25% compared with grinding and rolling. Figure 5 further illustrates that the relationship between the two methods of measuring rumen digestion of starch were comparable.

The relationships among intake, site of digestion, and extent of starch digestion in dairy cattle were examined by combining the results of 14 production studies (8, 9, 16, 36, 37, 59, 60, 78, 83, 99, 107, 117, 130, 134). All studies utilized cows in the first half of lactation, having as the primary objective to compare NSC sources. Diet composition, DMI, and production performance data were reported. When RDS values were not reported, those in Table 2 were used. When not reported, rumen

degradable carbohydrate values were those reported by Nocek and Russell (97), postrumen digestibilities of starch were as in Table 7, or a value of 65% (101) was assumed. Specific intake parameters associated with site of digestion were compared to determine their degree of relationship (Table 5). Regression analyses were conducted to determine the presence of linear, quadratic, or cubic functions.

Total starch intake (kilograms per day) was correlated positively to intake (kilograms per day) levels of RDS, ruminal escape starch, total tract digestible starch, and intestinal digestible starch (IDS). Rumen degradable starch and rumen degradable carbohydrate [calculated by Nocek and Russell (97)] were correlated positively, as expected. However, RDS only accounted for about 60% of the variation associated with the rumen degradable carbohydrate component, indicating that structural carbohydrate digestion can contribute a large portion of the fermentable carbohydrate. Ruminally and intestinally digestible starch were corre-

lated in a positive fashion. Ruminal and total tract digestibility of starch were highly related. Also, IDS and total tract digestibility were related positively. Although this relationship was not as strong as that between RDS and total tract digestible starch, total tract digestible starch is influenced to a greater degree by ruminal rather than by postruminal digestibility.

The relationship between starch escaping the rumen and intestinal digestion was evalu-

ated to determine whether postruminal starch digestion was limiting. There was a positive relationship, indicating that, as ruminal escape starch increased, intestinal digestion also increased (Figure 6). No quadratic or cubic components were identified in this relationship. It must be borne in mind that intestinal digestion represented both small and large intestines. These data compare with those of Owens et al. (101), who demonstrated a similar effect in a summary of beef cattle data.

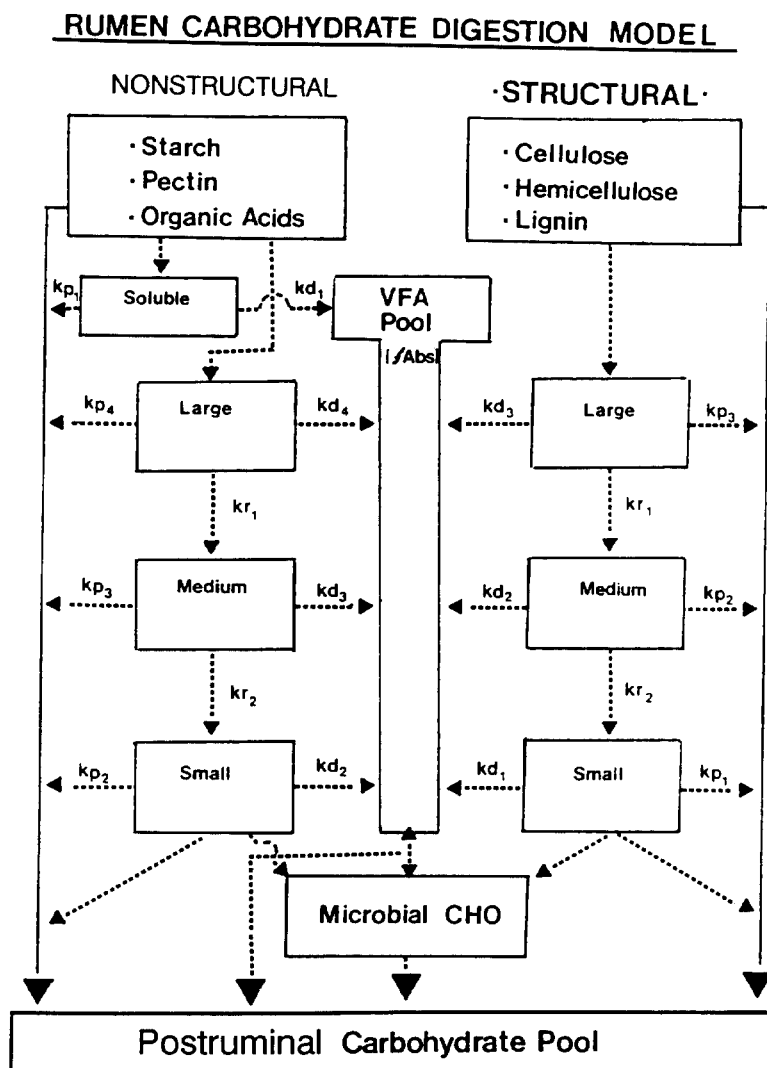


Figure 3. Rumen carbohydrate (CHO) digestion model; kd = rate of digestion, kp = rate of passage, kr = rate of particle size reduction, $1/Abs$ = a function of absorption. Large, medium and small refer to relative particle sizes.

Although more starch is digested postruminally as its intake is increased, efficiency of starch digestion suffers. Starch entering the small intestine (percentage of intake) was plotted against intestinal starch digestion (percentage entering) (Figure 7). The regression showed that, as starch delivery increased, starch digestion as a percentage of that entering the small intestine decreased. This relationship coincides with data demonstrating decreased energy digestibility associated with increased intake above maintenance (12, 84).

Small Intestinal Carbohydrate Uptake and Glucose Absorption

Glucose uptake by the mammary gland is the major determinant of rate of milk secretion (63). Mammary glucose uptake is a linear function of glucose availability to the gland, as is the relationship between milk production and plasma glucose concentration (63, 64). Kronfeld et al. (65) calculated that 72 g of glucose uptake by the mammary gland were required to produce 1 kg of milk. Thus, a cow producing 30 kg of milk requires about 2.2 kg/d of glucose uptake by the mammary gland. With this estimated requirement for glucose, dietary manipulation to enhance supply of glucose to the mammary gland may be a useful strategy to increase production performance.

In the ruminant, logical modes of increasing plasma glucose are through providing rumen fermentable carbohydrates that result in glucogenic substrates for the liver and starch sources

that escape ruminal fermentation. These approaches should result in a net positive glucose appearance at the portal-drained viscera (PDV). In theory, the latter should be an energetically efficient means to provide increased circulating glucose (less heat loss due to fermentation), providing that ruminally available carbohydrate does not limit microbial protein synthesis. The way in which the small intestine digests and absorbs starch and the resultant glucose available for mammary uptake is influenced by several factors: 1) intestinal carbohydrase and pancreatic amylase activities and starch hydrolysis, 2) visceral metabolism of glucose, and 3) regulation of endogenous glucose.

Enzymatic Activity. Owens et al. (101) indicated that limited enzymatic activity for starch hydrolysis and adequate access of enzymes to starch granules are factors limiting intestinal starch digestion in ruminants. Two primary sources of postruminal carbohydrase activity are the pancreas and intestinal mucosa. Intestinal breakdown of starch to glucose requires enzymes capable of cleaving both α -1-4 and α -1-6 glucosidic linkages (amylose and amylopectin). Pancreatic amylase breaks down amylose to maltotriose and maltose. Maltase activity in the small intestine yields glucose (13). Hydrolysis of amylopectin is completed in the small intestine by isomaltase, because pancreatic amylase has no α -1-6 glucosidic activity (125).

Kreikemeier et al. (62) fed calves 90% forage or 90% concentrate diets at either once or

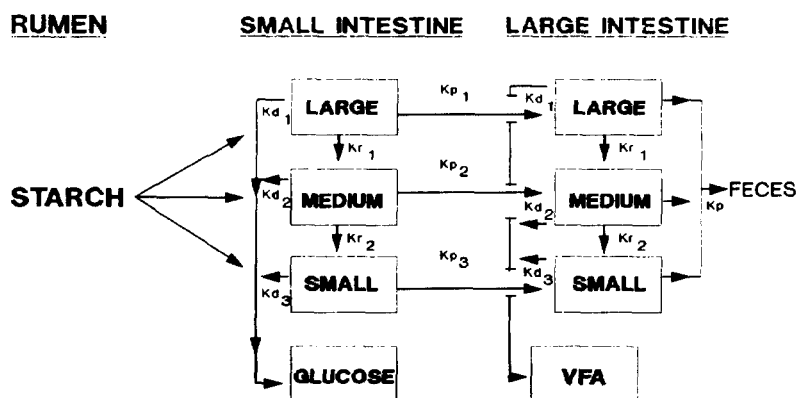


Figure 4. Postruminal digestion model. Large, medium, and small refer to particle sizes within a compartment; k_d = rate of digestion, k_r = rate of particle reduction, k_p = rate of particle passage.

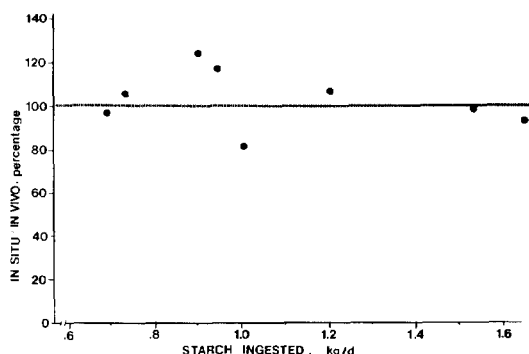


Figure 5. Relationship between flow of starch originating from a mixture of sources estimated from nylon bag studies (in situ) and measured in vivo in dairy cows at different starch intakes (each point is the mean of 8 nylon bag measurements and 1 to 3 cow measurements in vivo).

twice net energy for maintenance and evaluated digestive enzymatic activity. Increasing feed intake increased pancreatic amylase and glucoamylase activities, small intestine length, digesta weight, and α -amylase activity. Calves fed forage had higher activities of these enzymes regardless of intake level. Russell et al. (120) showed that steers fed hay had higher (44%) pancreatic amylase activity than those fed corn at the same level of intake. Increasing energy intake to three times maintenance resulted in 84% increase in pancreatic α -amylase activity. On the other hand, Janes et al. (54) reported a 34% increase in pancreatic α -amylase activity when sheep were fed corn compared with dried grass diets. Others sup-

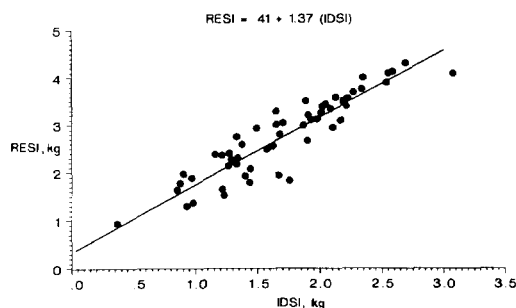


Figure 6. Relationship between rumen escape starch intake (RESI) and intestinally digested starch intake (IDSI) in dairy cattle ($r^2 = .82$, $P < .001$).

port this finding (10, 142). Short-chain VFA have been shown to stimulate pancreatic amylase secretion (58). These workers demonstrated that pancreatic amylase release in vitro was increased by increasing concentration (10^{-5} to 10^{-2}) of butyrate. Isovalerate demonstrated the greatest release response. These data suggest that regulation of pancreatic amylase secretion is influenced by both level of feed intake and diet composition.

Increased dietary protein concentration was shown to increase starch digestion (121, 146). Fuskiki and Iwai (28) hypothesized (Figure 8) that high feed protein intake spares cholecystokinin-releasing peptide from digestion by trypsin. This allows the peptide to release cholecystokinin from the intestinal mucosa, thus stimulating amylase release.

There are three primary intestinal carbohydrases identified as contributing significantly to starch digestion: isomaltase, maltase, and lactase (62). Isomaltase and maltase activities appear to be greatest in the midjejunum and ileum (13, 62). Diet has little influence on the distribution or activity of carbohydrases in the small intestine (54, 120). However, calves fed forage had higher activities than those fed high grain diets, whereas increased energy intake elevated activities of all intestinal carbohydrases in cattle (62).

It has been suggested that capacity for intestinal starch digestion in the ruminant may be

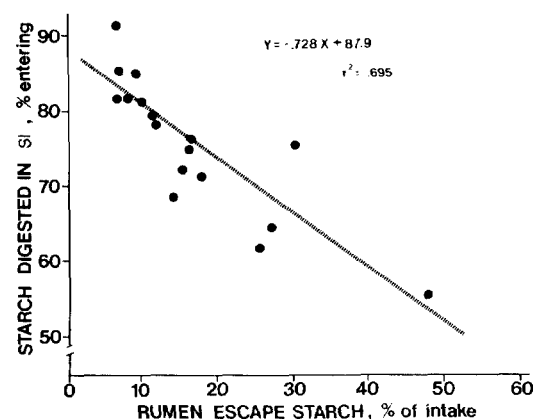


Figure 7. Relation between the resistance of starch against rumen degradation (percentage escape) and digestion in the small intestine (SI) of dairy cows.

limited by enzymatic hydrolysis, in particular, low amylase and inadequate isomaltose activity (75). Mayes and Orskov (75) demonstrated that the capacity of small intestine of sheep to digest and absorb raw starch was 100 to 200 g/d, compared with 200 to 300 g/d for gelled starch and 300 to 400 g/d for glucose. They attributed this lower digestion of raw starch to limited pancreatic amylase secretion and short retention times of starch. It also was suggested that the capacity for glucose absorption may limit starch digestion. However, infusion of amylase in steers did not increase starch digestion (108), suggesting that amylase activity may not limit intestinal starch digestion.

Glucose Absorption and Metabolism by Viscera Tissue

Amount of starch escaping the rumen is determined by extent of ruminal digestion. Depending on a variety of conditions (grain, processing, and diet), rumen escape starch can exceed 50% of the dietary starch intake in dairy cattle (37, 78). If this starch were digested and absorbed by the small intestine as glucose, it could contribute significantly to the total glucose pool. There is some question, however, as to what proportion of starch entering the small intestine actually is absorbed as glucose and, once absorbed, what its metabolic fate is upon delivery to the hepatic portal vein.

Most studies in dairy cattle have been limited to *in vivo* measurements to determine starch disappearance from the gastrointestinal tract. These studies do not consider the cost of metabolism by gut tissue, which ultimately influences net presentation of nutrients (net absorption) to the liver (portal vein) and other tissues (muscle and mammary gland) (63, 113). The use of chronic indwelling catheters appropriately placed in blood vessels can render useful information regarding metabolism of these nutrients in a given tissue. It is important in evaluating this type of research that the exact locations of cannulas are identified in order to define specific tissues or organs representing net fluxes. Figure 9 illustrates a general overview of site digestion and metabolite sources.

Several studies (47, 49, 68, 110) indicated no net glucose absorption from the hepatic PDV in dairy cattle fed normal lactating diets.

This implies limited uptake of glucose from dietary sources (limited intestinal hydrolysis, transport) or that absorbed glucose is used for gut metabolism.

Net Glucose Absorption. Huntington and Reynolds (48) evaluated the effect of abomasal infusion of water, glucose, or corn starch on net absorption of various metabolites by dairy cows fed a 60% corn silage and 40% concentrate diet. Net absorption of glucose at PDV was -56, 34, and -46 mmol/h for water, glucose, and starch infusions, respectively. For beef heifers, portal appearance of glucose was 78 and 27.4 mmol/h for glucose and starch infusions, respectively. Calculated recovery of infused starch as glucose in dairy cows was 8% compared with 35% in beef heifers, whereas recovery of infused glucose as glucose was similar between dairy cows and beef heifers (64 and 65%). These data suggest limitation to starch hydrolysis and glucose absorption at the gut level or specific differences in the way that dairy and beef cattle utilize glucose in the intestine. Beef heifers were fed either alfalfa hay or high grain (restricted or for *ad libitum* intake). No differences were seen in the recovery of absorbed or infused glucose among diets.

Kreikemeier et al. (61) infused 0, 20, 40, or 60 g/h of glucose, corn starch, or corn dextrin abomasally into steers. As the concentration of these infused compounds increased, so did the amount of carbohydrate disappearing from the small intestine. A linear increase in net glucose appearance in PDV was associated with increased glucose infusion. However, as increasing levels of starch were infused, net portal appearance of glucose maximized at the 20 g/h infusion rate (Figure 10). Because unpolymersized glucose was present at the ileum with starch infusion, the authors concluded that starch hydrolysis exceeded the capacity for glucose absorption.

Janes et al. (55) fed sheep dried grass or corn-based diets that yielded 47 and 530 g of dietary α -glucoside. The relative proportions of α -glucoside digested in the rumen, small intestine, and large intestine were similar between diets. As expected, approximately seven times as much starch entered the small intestine for the sheep fed corn. There was a positive net glucose absorption for sheep fed corn (19.28 mmol/h) compared with a slightly nega-

TABLE 4. Starch escaping degradation in the rumen of dairy cows, measured with nylon bag incubation studies.¹

Ingredient	Starch	Soluble ²	Kd ³	Ruminal escape starch		
				1 ⁴	2 ⁵	3 ⁶
	(% of DM)	(% of total starch)	(%/h)	(% of total starch)		
Barley 1	57.2	47.0	15.4	13.0	17.7	13.3
Barley 2	56.1	62.0	24.2	6.5	12.7	9.5
Beans (<i>Vicia faba</i>) 1	42.2	41.0	12.2	17.2	21.3	15.9
Beans (<i>Vicia faba</i>) 2	36.0	44.0	8.0	21.5	25.9	19.5
Beans (<i>Phaseolus</i>)	34.7	14.0	10.0	28.7	30.1	22.6
Corn 1	68.1	19.0	4.0	45.0	46.0	35.2
Corn 2	67.6	27.0	4.0	40.6	43.3	32.4
Corn 3	65.2	32.0	3.6	39.5	42.7	32.1
Corn gluten feed 1	40.3	70.0	7.8	11.7	18.7	14.0
Corn gluten feed 2	24.6	50.0	12.6	14.2	19.2	14.4
Corn gluten feed 3	18.9	55.0	10.2	14.8	20.3	15.2
Corn gluten meal	20.5	23.0	28.6	11.5	13.8	10.3
Hominy feed 1	60.7	30.0	4.9	35.4	38.4	28.8
Hominy feed 2	33.6	48.0	5.6	24.5	29.3	22.0
Millet	61.6	41.0	8.3	22.2	26.3	19.7
Milo	65.2	32.0	3.5	39.5	42.7	32.1
Oats	46.6	96.0	18.8	.8	10.4	7.8
Peas 1	42.6	24.0	14.1	19.9	22.3	16.7
Peas 2	41.8	50.0	7.1	20.7	25.7	19.2
Potatoes	65.1	26.0	4.9	37.4	40.0	30.0
Potato starch	65.0	23.0	16.0	18.3	20.6	15.5
Rice	84.8	26.0	7.6	29.4	32.0	24.0
Rice bran 1	31.7	14.0	15.4	21.1	22.5	16.9
Rice bran 2	26.9	34.0	9.3	23.1	26.5	19.9
Rice bran 3	18.0	0	13.5	27.0	27.0	20.3
Rice bran 4	22.1	23.0	8.9	27.7	30.0	22.5
Tapioca 1	72.6	75.0	16.8	5.7	13.2	9.9
Tapioca 2	71.0	68.0	13.1	8.8	15.6	11.7
Tapioca 3	68.4	59.0	6.8	17.4	23.3	17.5
Wheat 1	66.6	68.0	18.8	6.7	13.5	10.1
Wheat 2	65.4	68.0	17.5	7.1	13.9	10.4
Wheat flour 1	77.2	87.0	19.8	2.6	11.3	8.5
Wheat flour 2	49.3	85.0	15.7	3.6	12.1	9.1
Wheat middlings 1	23.8	80.0	21.1	3.8	11.8	8.9
Wheat middlings 2	20.0	89.0	27.2	1.7	10.6	8.0
Wheat bran	13.4	81.0	20.8	3.7	11.8	8.8

¹Information from Tamminga et al. (135).²Percentage of starch washed out of nylon bags without rumen incubation.³Kd = Rate of disappearance of starch from nylon bags suspended in the rumen.⁴1 = Arbitrarily assumed that only insoluble starch can escape.⁵2 = Arbitrarily assumed that 10% of soluble starch escapes rumen degradation.⁶3 = Escape values of ground starch sources corrected for pelleting (reduced by 25%).

tive absorption (-2.03 mmol/h) for sheep fed grass. Peripheral blood glucose concentrations were similar for both diets. Weighart et al. (153) demonstrated greater positive net portal appearance of glucose in lactating cows fed a high concentrate diet for ad libitum intake compared with its restricted feeding (138 vs. 60 mmol/h). However, no differences were

detected between high concentrate and high forage diets fed at equal energy intakes. They conducted regression analyses on beef and dairy cattle data and showed a positive portal appearance of glucose when daily digestible energy intake (DEI) was greater than 23 Mcal/d. As DEI increased beyond this concentration, net gut plus hepatic glucose outputs increased.

TABLE 5. Relationship between specific carbohydrate site digestion parameters in dairy cattle.¹

Variables		Equation components		
Dependent	Independent	n	r	Intercept x x ²
Starch intake	Rumen degradable starch (RDS)	59	.92	1.54 1.26 . . .
Starch intake	Rumen escape starch (RES)	59	.71	3.32 2.38 . . .
Starch intake	Intestinal digestible starch (IDS)	59	.98	-.07 .87 . . .
Starch intake	Total tract digestible starch (TTDS)	59	.77	.26 .34 . . .
RDS	Rumen degradable carbohydrates	59	.61	6.34 .81 . . .
RDS	IDS	59	.43	53.59 .51 . . .
RDS	TTDS	59	.95	-1.57 1.22 -.036
RES	IDS	59	.91	.41 1.37 . . .
IDS	TTDS	59	.68	

¹References used to develop relationship: (8, 9, 16, 36, 37, 59, 60, 78, 83, 99, 107, 117, 130, 134).²All correlations were significant ($P < .01$).

Reynolds et al. (114) demonstrated an increase in net removal of glucose by PDV with higher levels of intake in diets with either 75% concentrate or 75% forage fed to heifers. Heifers fed concentrate had a higher relative non-significant difference than those fed forage; however, those fed concentrate had a positive glucose absorption at low intake (37 mmol/h) compared with a negative absorption for heifers fed forage (-8 mmol/h).

Reynolds and Huntington (110) fed steers either alfalfa or high concentrate-based diets. With alfalfa-based diets, glucose in mesenteric-drained viscera (MDV) was 69% of PDV use. Concentrate diet MDV showed a net absorption (29 mmol/h), reducing net PDV use to 1 mmol/h. This suggested that any net gain of glucose via intestinal absorption was being utilized primarily for stomach metabolism

(PVD - MDV). They concluded that differential use of metabolizable energy for tissue energy in heifers fed different forage to concentrate ratios and energy densities were associated with changes in PDV heat production. The greater use of PDV oxygen for forage compared with concentrate diets accounted for over 50% of the difference in maintenance energy cost. The difference in the metabolic activity of visceral tissues may be associated with differences in work of digestion, absorption, and the components of metabolizable energy absorbed (114).

There appears to be a threshold at which a net glucose appearance at the portal vein is observed. This seems to be evident either when glucose is infused or when intakes are high, both of which promote increased starch or DEI and rumen bypass of starch. Density and intake

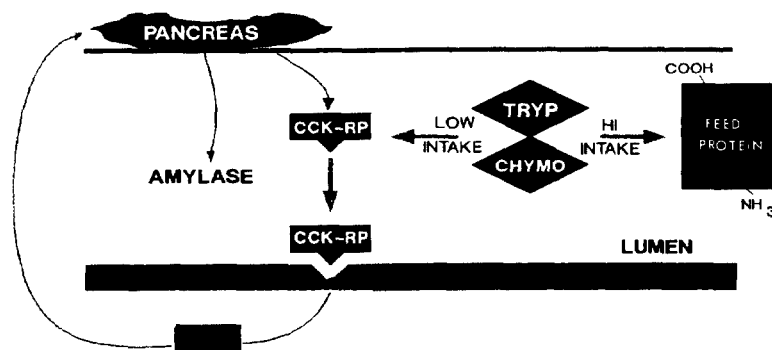


Figure 8. Proposed mechanism for pancreatic amylase release as influenced by dietary protein concentration (28); cholecystokinin pancreozymin-releasing peptide (CCK-RP), trypsin (TRYP), chymotrypsin (CHYMO).

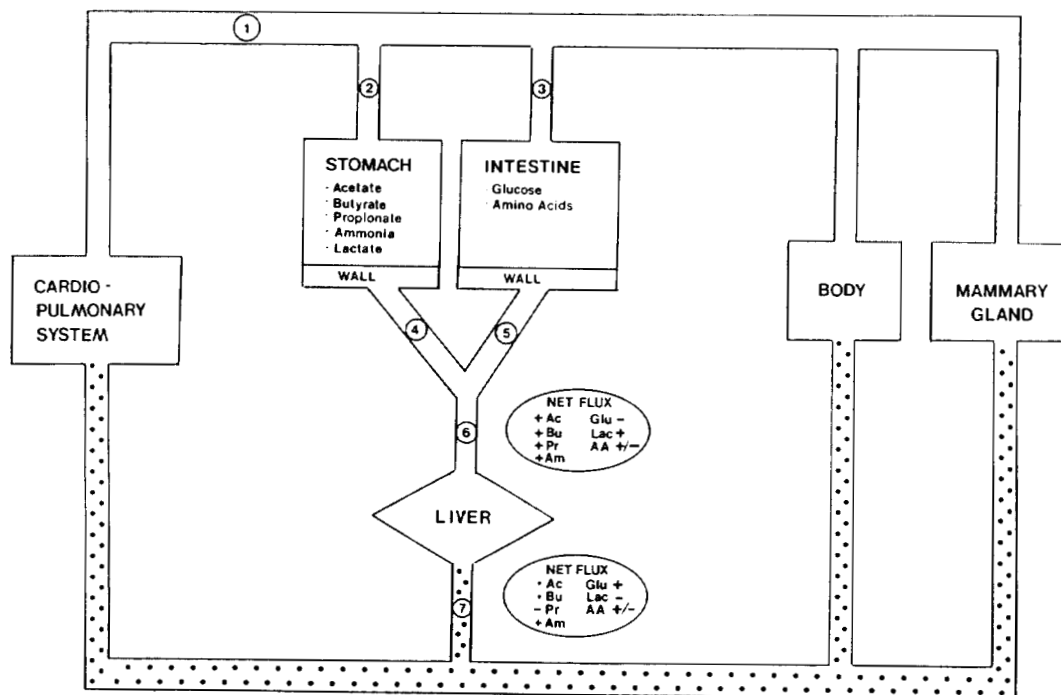


Figure 9. Illustration of general site digestion, blood flow, metabolite source and flux in dairy cows. The dotted area represents venous blood. Sites of catheterization: 1) carotid artery, 2) gastrosplenic artery, 3) mesenteric artery, 4) gastrosplenic vein, 5) mesenteric vein, 6) portal vein, and 7) hepatic vein.

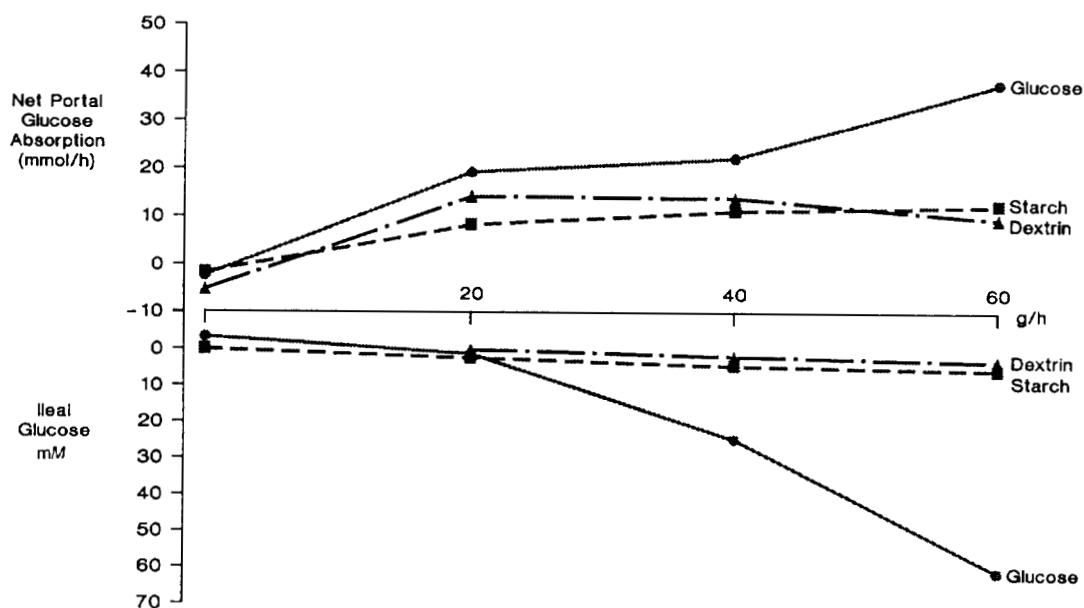


Figure 10. Relationship between ileal glucose (mM) and net portal appearance of glucose (mmol/h) at different levels of abomasally infused glucose, starch, and starch dextrin: 0, 20, 40, and 60 g/h (61).

of metabolizable energy also dictate the relative glucose needed to support metabolism of PDV. It seems this requirement must first be met prior to net positive absorption of glucose at the portal vein.

Visceral Glucose Metabolism. In sheep fed corn or dried grass-based diets, lactate utilization was equal to rate of glucose production; diet did not influence the proportion of lactate derived from intestinal glucose metabolism (55). Up to 42.5% of the L-lactate appearing in MDV originated from glucose metabolism, suggesting considerable glycolytic activity in the small intestine. Measurements of metabolism in MDV demonstrated that net increase in lactate and propionate absorption tended to be higher for steers fed high alfalfa compared with high concentrate diets (111). The stomach and intestine represented 85 to 100 and 75% of net VFA and lactate appearance, respectively. Arterial glucose and propionate concentrations were not influenced by diet when animals were meal-fed. Conversion of gut-derived L-lactate from propionate via gut metabolism has not been determined (111). Lactate carbon can be converted to glucose and other metabolites in the liver (112). However, lactate carbon accounts for only 12 to 15% of whole body glucose (4, 141).

Acetate utilization by PDV in sheep (104) is associated primarily with oxidative metabolism rather than incorporation into mesenteric fat and represents about 25% of the whole body acetate turnover on alfalfa-based diets. Propionate is derived primarily from gut absorption and shows a net positive absorption in PDV. Hepatic uptake is associated primarily with glucogenesis.

In lactating cows (4 and 8 wk postpartum), net PDV glucose flux was negative and not influenced by week postpartum (110). This negative flux was a result of greater use of arterial glucose by PDV rather than absorption from the gut into portal blood (55). These cows were fed diets consisting of 60% corn silage and 40% concentrate. A higher level of concentrate, as may be appropriate in this stage of lactation, may have resulted in a greater net flux of PDV glucose. Nonetheless, net hepatic glucose production (3.1 kg/d) more than adequately met mammary (2.3 kg/d) and other glucose requirements of cows producing 32 kg of milk (111). Lomax and Baird (68) demon-

strated net utilization of glucose (-2 mmol/min) across the gut of lactating dairy cows. Lactate, propionate, and acetate net utilization in PDV were positive, indicating net output. Hepatic output demonstrated a net synthesis of glucose (5.41 mmol/min) with net uptakes of lactate, propionate, acetate, butyrate, and glycerol of -1.73 , -4.98 , $-.45$, $.06$, and $-.09$ mmol/min, respectively. Net hepatic uptake of glucogenic precursors could account for only 72% of the hepatic output of glucose. Fasting for up to 6 h decreased hepatic output of glucose by 77% and increased uptake of pyruvate, glycerol, and NEFA. Hepatic output of glucose was correlated highly with DE intake and milk yield. In fed cows, the most important precursor to glucose was propionate, potentially accounting for 50% of the hepatic glucose output. Fasting caused the contribution of propionate to fall to insignificant levels; other glucogenic precursors (derived endogenously) accounted for all hepatic glucose output. Table 6 summarizes metabolic net fluxes across the PDV and hepatic tissues in dairy cattle in different stages of lactation.

Given the relationship between starch intake, site of digestibility, and extent of digestion, the question remains whether manipulating site of digestion will influence substrate provided for use in production. Several studies employing PDV or MDV catheterization techniques with dairy cattle were compiled to evaluate the relation of digestible energy, RDS, and IDS intake on appearance of glucose, propionate, and lactate in PDV. The data represented eight studies with 27 treatments with nonlactating cows and cows producing up to 32 kg/d of milk. Simple correlation coefficients were computed between selected measures of carbohydrate intake and net glucose and propionate appearance at the portal vein.

Table 7 shows a negative relationship between DEI and carbohydrate intake versus net portal appearance of glucose. Propionate and lactate, however, showed a strong positive relationship with intake parameters. The data of Weighart et al. (153) were the only ones that consistently demonstrated a positive appearance of glucose from the PDV for high producing dairy cows. These researchers developed a data base consisting of steers and dry and lactating cows (23 observations). Regressing DEI on net PDV glucose flux resulted in a

TABLE 6. Net flux of metabolites across portal drained viscera (PDV) and hepatic tissue in dairy cattle at different stages of lactation.¹

Metabolite	Tissue	Flux	Comment
Acetate	PDV	Positive	Reflects NE _L and rumen degradable starch intake. Used by liver as oxidative carbon units.
Propionate	Hepatic	Net removal	Can spare glucose for milk synthesis.
	PDV	Positive	Degree reflects NE _L and rumen degradable starch intake.
	Hepatic	Net removal	Major glucose precursor. Glucose synthesis accounts for 95% of propionate metabolism.
Lactate	PDV	Positive	Reflects gut tissue metabolism or absorption from gut.
	Hepatic	Net removal	Used by tissue to synthesize glucose and other metabolites. Accounts for 10 to 12% of whole body glucose production.
Glucose	PDV	Negative	Reflects greater use of arterial glucose by PDV than absorption of glucose from the gut into the portal vein.
	Hepatic	Net production	Reflects energy intake and milk production.

¹Summary of nonlactating and lactating dairy cows in wk 4 and 8 postpartum (68, 109, 112); the magnitude of metabolite flux varied with stage of lactation; however, the direction of flux remained constant.

positive relationship ($r = .42$, $P < .05$). After further aggregation of the data into energy levels, the following equation was developed: PDV glucose (mmol/min) = $-11.5 + .54 \times \text{DEI}$, $r = .98$. This relationship suggests that when DEI exceeds 23 Mcal/d, glucose appearance at the portal vein will be positive. Figure 11 illustrates the relationship between DEI and PDV glucose when regression analyses on the data were conducted with and without the data of Weighart et al. (153). Although trends were found, only 3 data points changed the direction of the relationship.

Possible explanations for the different relationship between DEI and PDV flux would be as follows: positive flux would indicate that a threshold of glucose is needed for gut metabolism, and, once met, a net positive portal appearance would occur. Negative flux would suggest that, as DEI increases, there is a greater oxidative drain on glucose for gut metabolism to process and metabolize nutrients (increased metabolic and secretory cell processes, protein turnover, cell transport, etc.). However, there probably are not enough data to develop a sound relationship or conclusion.

Regulation of Glucose Metabolism. Because plasma glucose levels and milk production are correlated (63, 64), increasing plasma glucose should be beneficial. Several studies (45, 66, 102, 103) have shown increased plasma glucose in response to abomasal glucose infusion. However, abomasal infusion of starch has elicited no response in blood glucose (45, 66,

103). Net nutrient absorption across the MDV determined in sheep (55) showed that when they were fed corn diets, whole body glucose turnover rate was greater than for grass diets (38.9 vs. 28.4 mmol/h). The contribution of absorbed glucose to glucose turnover rate also was considerably greater for sheep fed corn versus grass (61 vs. 8%). However, endogenous glucose production was lower for corn-based diets (15.6 vs. 26.9 mmol/h). The sparing effect of endogenous precursors by exogenous glucose absorption on high corn diets was not a 1:1 ratio. Thus, a greater whole body glucose turnover rate would be expected. Although PDV glucose appearance can be positive, it does not mean that plasma levels will be increased. The balance between exogenous provision and endogenous synthesis of glucose indicates a highly regulated system.

The fact that no net portal appearance of glucose has been demonstrated and that plasma glucose concentrations are not elevated with starch infusions may be associated with the short-term nature of infusion studies. Kreikemeier et al. (61) found a two- to threefold increase in net hepatic glucose appearance with glucose compared with starch at different levels of postprandial infusion. These steers were fed a basal diet of alfalfa hay with a 3-d noninfusion period between substrate infusions. Studies to evaluate net glucose absorption at the portal vein thus far have been of a rotation nature in design, with a 2 to 3-wk period or less for ration adaptation. It may be

TABLE 7. Simple correlations between energy and carbohydrate intake and net portal appearance of metabolites.¹

Intake variable, kg/d	Metabolites, mmol/h		
	Glucose	Propionate	Lactate
DMI	-.43 ^a	.93 ^c	.77 ^c
Digestible energy intake, Mcal/d	-.42 ^a	.94 ^c	.77 ^c
Starch intake	-.37 ^b	.94 ^c	.77 ^c
Rumen degradable starch intake	-.37 ^b	.94 ^c	.77 ^c

^{a,b,c} $P < .10$, $P = .11$, $P < .001$.

¹References (46, 47, 48, 49, 68, 109, 112).

that this time period in relation to basal diet feeding or substrate infusion is not adequate for intestinal or pancreatic enzymatic adaptation or transport site activation for these different substrates to be evaluated fairly.

Insulin. Orskov (100) concluded that although hepatic tissues have a great capacity to synthesize glucose, this process is an inefficient utilization of exogenous glucose. When portal propionate concentrations exceed the capacity of the liver to utilize it, insulin production is stimulated. The result in lactating cows is a shift from tissue lipolysis to lipogenesis, resulting in decreased milk and fat yields. This interpretation is based partly on work by Istasse and Orskov (53), who showed that intermittent infusion of propionate at a rapid rate into the rumen of lactating dairy cows caused an immediate increase in insulin production. A continuous rate of propionate infusion had no effect on insulin release.

In lactating dairy cows fed hay and concentrate separately according to milk production, Lomax and Baird (68) found the difference in portal hepatic exchange of propionate to be .34 mmol/h. There was no indication that this would cause a dramatic increase in insulin release. When sheep were infused ruminally with a basal level of VFA plus propionate, glucose, or the combination, increased propionate infusions caused increased PDV insulin flux but had no effect on arterial insulin (33).

Janes et al. (55) evaluated insulin action and glucose metabolism in sheep fed dried grass or corn-based diets. Peripheral glucose concentrations were not affected by diet. They showed a difference due to diet in maximum metabolic clearance rate of glucose, which suggests an increased responsiveness to insulin in glucose utilization for sheep fed corn diets. However,

they attributed part of the difference to a change in rate of glucose disposal not mediated by insulin. Plasmin insulin concentrations required to suppress endogenous glucose production were greater than those reported for humans. They concluded that increased utilization of glucose in sheep fed corn was not mediated by a change in the sensitivity to insulin and only partially due to a responsiveness in insulin. They also indicated that plasma insulin does not play a major role in controlling endogenous glucose production in sheep.

Accountability of Small Intestinal Carbohydrate Disappearance

Total accountability for α -glucoside that is presented to the small intestine is a challenge. Janes et al. (55) accounted for 83 to 111% of the starch disappearance in the small intestine as net glucose absorbed in the MDV from grass and corn diets, respectively. In another study (48), recovery of abomasally-infused glucose as absorbed glucose in the PDV was 65%. However, when only starch was infused, 35 and 8% were accounted as absorbed glucose in heifers and in a cow, respectively. Kreikmeier et al. (61) accounted for 94% of abomasally-infused (60 g/h) glucose as net glucose absorbed. However, only 66 and 44% of starch and dextrin could be accounted for as absorbed glucose. Major areas affecting glucose accountability could include glucose transport and its microbial fermentation in the small intestine.

Glucose Transport in the Small Intestine. Unlike monogastric animals, ruminants ingest large amounts of carbohydrates that are digested in the rumen, and variable amounts reach the small intestine. Rate of glucose ab-

sorption in young lambs is greater than in adult sheep (155). In addition, as the ruminant is weaned to dry feed, transport capacity decreases as the rumen develops. Thus, the capacity of ruminants to transport glucose across the intestine may be limited.

Intestinal glucose absorption can occur either as passive (first-order) or active (zero-order) transport (155). In adult sheep, the absorptive process can be described by first-order kinetics. Approximations of intestinal glucose absorption are grossly underestimated when zero-order kinetics are used. In young lambs, the rate of glucose transport decreased linearly with distance from the pylorus. This occurrence was not as pronounced in adult sheep.

In ruminants, the mechanism of D-glucose transport across the small intestine is by active, sodium-dependent pump (Figure 12). Scharrer (123) indicated that a low capacity system of sodium-dependent sugar transport existed in sheep as a result of finding that sugars were transported against a small concentration gradient to a higher sugar concentration. Thus, active transport in the ruminant occurs by a high affinity, low capacity system as opposed to the pig in which a low affinity, high capacity system operates (157). The lower D-glucose transport capacity in sheep intestine (jejunal) compared with the pig probably reflects fewer carrier molecules in the brush border membrane of the former. This system (high affinity, low capacity) favors absorption at low intestinal glucose concentrations.

Given the transport capabilities of the adult ruminant, one must evaluate whether the gut has a limited capacity to transport glucose. Net glucose absorption at PDV and unpolymerized glucose flow to the ileum increased with increased levels of abomasally-infused glucose and starch (61). Moreover, capacity of starch and dextrin hydrolysis in the small intestine that could be accounted for as net glucose absorption was maximized at 20 g/h infusion rate. The fact that concentration of unpolymerized glucose in ileal digesta increased as glucose concentration increased suggests that hydrolysis occurred at a rate faster than transport. These workers (61) also indicated that, when 20 g/h of glucose were infused, 1.1 mmol of glucose remained in ileum digesta. This strongly suggests saturation of the active transport system or some other barrier (physical or chemical) to glucose transport. However, at

higher levels of infusion, passive transport appeared to predominate, because glucose absorption rose slightly with infusion rate. It was suggested that starch hydrolysis occurred at a faster rate than glucose transport in distal segments of the small intestine. The authors indicated that these data are consistent with the calculations of Owens et al. (101), which suggest that there is no plateau in the amount of starch disappearing in the small intestine.

The transport process is energy demanding. McBride and Milligan (76) indicated that maintenance of Na and K gradients across the plasma membrane in mucosal cells is an energetically costly function. The maintenance of respiration dependent on Na-K-ATPase accounts for 28 to 60% of in vitro oxygen consumption in duodenal mucosal cells. This expenditure increases with increasing DEI. In the rat, utilizing acute in vivo studies, Rich-Denson and Kimura (115) indicated that the intestine was not a significant source of portal-venous lactate, suggesting that glucose demanded for mucosal energetic purposes was relatively small. This is supported by Windmueller and Spaeth (156), who showed that 97% of luminal glucose was absorbed intact across the intestinal wall of fed rats. Because the active transport capabilities of the ruminant generally are less than those of monogastric animals, it would appear that the transport process requires little glucose.

Microbial Fermentation in the Small Intestine. A major route of intestinal glucose disap-

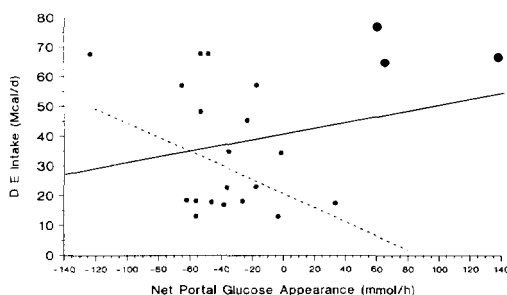


Figure 11. Relationship between digestible energy intake [(DEI), Mcal/d] and net portal appearance of glucose [(Glu), mmol/h] in dairy cattle (46, 47, 48, 49, 68, 109, 112, 153). Diamonds are the data points of Weighart et al. (153). Dotted line is the regression line without the data of Weighart et al. (153). (DEI = $24.6 - .21 \times (\text{Glu})$; $r = -.42$, $P = .07$). Solid line includes the data points of Weighart et al. (153) (DEI = $41.4 + .10 \times (\text{Glu})$; $r = .22$, $P = .28$).

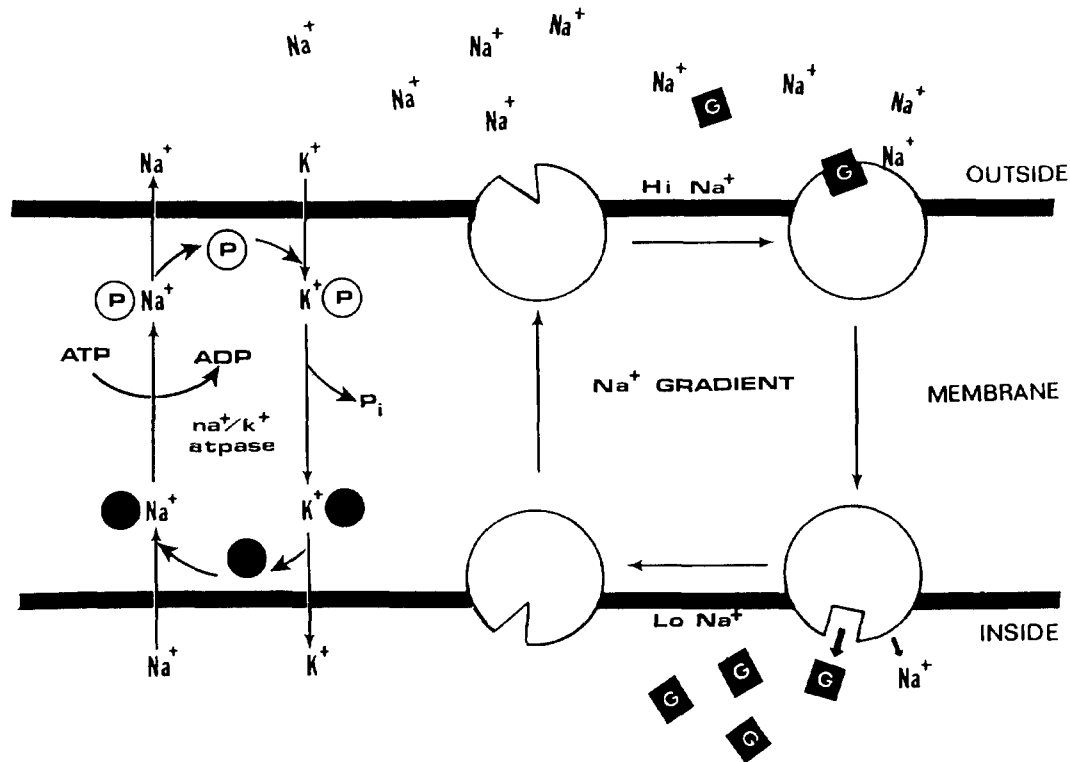


Figure 12. Illustration of the Na-K-dependent, ATPase active transport system for glucose (G) in the intestinal mucosa.

pearance not accounted for may be associated with fermentation by microbes inhabiting the small intestine. Nicoletti et al. (90) determined aerobically in sheep intestine that viable counts of microbial organisms ranged from 5×10 to 6×10^4 /g and for digesta from 10^4 to 5×10^6 cells/g. They also indicated that indigenous, acid-tolerant bacteria were able to ferment sugar and starch. Kreikemeier et al. (61) showed a linear increase in acetate and total VFA flow and a linear decrease in ileal digesta pH with increased levels of dextrin infusion. These data suggest the presence of small intestinal microorganisms capable of fermenting starch. However, the authors indicated that VFA concentration of PDV did not account for the deficit in glucose infused versus that absorbed. It has not been determined to what degree these microorganisms compete with the host animal for glucose or whether their fermentation can account for a significant proportion of the unaccountable glucose.

Effect of Site of Carbohydrate Digestion on Production Performance

Production performance is a major indicator as to whether or not manipulating site of carbohydrate digestion is useful. Although production studies in the past have evaluated ingredient effects in dairy cattle, only within the past few years have researchers utilized techniques to aid in evaluating influence of digestion site on performance. To date, most studies have evaluated ingredient effects with the basic premise that concepts such as RDS, escape starch, and carbohydrate degradability would follow. There may be inherent factors that will not allow specific ingredients to conform to a conceptual framework.

Table 8 shows the effect of starch source on milk production and composition in several production studies. The magnitude of milk yield response was variable among studies. Changes in milk composition were mainly in fat content. Only after feeding oats instead of

TABLE 8. Milk production results after feeding different starch sources.

Source ¹	Forage	Concentrate		Starch source	Cow per treatment	Milk	Fat	Protein	Reference
	Intake	Source	Intake						
	(DM, kg/d)		(DM, kg/d)	(% of concentrate)		(kg/d)	— (%) —		
Alfalfa hay	8.9	Wheat	7.9	96.2	5	23.3	3.93	3.69	(139)
	8.8	Corn	7.6	96.1	5	23.9	3.83	3.53	
	8.9	Milo	7.7	96.1	5	23.7	3.77	3.66	
	8.9	Oats	8.2	95.1	5	23.2	3.14	3.45	
	8.9	Barley	7.9	96.2	5	23.5	3.97	3.58	
Grass hay	6.0	Barley	9.0	82.2	16	16.1	4.49	3.15	(133)
	6.0	Corn	9.0	77.8	16	18.9	4.04	3.00	
	1.4	Barley	12.9	86.8	16	20.6	2.06	3.03	
	1.4	Corn	12.9	82.9	16	15.6	2.97	3.43	
Alfalfa hay	9.2	Barley	9.2	80.4	8	27.4	2.81	3.23	(16)
	9.2	Corn	9.2	65.2	8	28.0	3.01	3.21	
Oat silage plus alfalfa hay	5.8	Barley	11.1	91.0	18	22.9	4.54	3.52	(85)
	6.2	Wheat	11.9	91.6	18	24.0	4.19	3.84	
	6.1	Oats	11.6	91.4	18	25.1	4.72	3.12	
Grass hay	3.6	Barley	8.9	64.0	8	15.9	4.20	3.39	(74)
	3.5	Oats	9.5	66.3	8	16.9	3.79	3.16	
Alfalfa silage	9.9	Barley	10.8	97.2	6	22.9	3.41	3.22	(154)
	14.2	Corn	7.0	88.6	6	23.3	3.75	3.20	
Corn silage plus alfalfa hay	9.4	Barley	9.4	70.2	20	31.6	3.36	3.00	(8)
	10.2	Corn	10.2	62.7	20	32.8	3.37	3.05	
Grass silage plus corn silage	12.2	Corn	12.2	50.0	16	37.1	4.34	3.55	(18)
	12.3	Barley	12.4	50.0	16	38.2	4.25	3.53	

¹Within each study, the same forage sources were fed among grain source treatments.

barley (74, 85) was milk protein content reduced slightly. The reason for this could be either the level of unsaturated fatty acids in oats or the presence of significant amounts of cell wall components.

McCarthy et al. (78) used lactating dairy cows to investigate the effect of protein (fish meal versus soybean meal) and carbohydrate source (corn versus barley) on ruminal and postruminal digestibilities and animal performance (Table 9). There were no significant ($P > .05$) percentage units higher for corn diets, and cows consumed more DM and, consequently, 21% more starch. Considerably more starch was delivered to the small intestine of cows fed corn versus barley diets. This suggests that a large amount of starch can be digested post-ruminally. Amount of digestion in the large intestine was not determined; however, it accounts for a relatively small percentage (1 to 8%) of starch in the diet (101).

Milk production was 3.1 kg higher for cows fed corn diets with 4% FCM only 1.1 kg higher (78). Protein percentage was not affected; protein yield followed that of milk. High energy intake appeared to be the main reason for increased production. Although site of starch digestion was shifted, it is difficult to identify the specific effects because energy intake and starch source were confounded. Apparently, the corn diets provided adequate ruminal degradable carbohydrate for microbial synthesis, because microbial N flow post-ruminally was not influenced by starch source.

Herrera-Seldana and Huber (37) fed cows in early lactation dry-rolled barley or milo with cottonseed meal or brewers dried grains as the major starch and protein sources. Starch and RDS intakes were highest for those fed barley and cottonseed meal and lowest for those fed milo and dried grains. This was largely due to the difference in starch concentrations in the

diets (32.8 vs. 24.4%, respectively). Regardless, starch delivery to the lower gut was similar among diets. The barley and cottonseed meal diet had the highest milk yields; other diets were similar but lower. Rumen degradable protein was 59.9 vs. 43.7% of CP on barley and cottonseed meal versus milo and brewers dried grains diets. The authors suggested a higher RDP in association with a high level of RDS would be needed to synchronize ruminal energy and protein utilization and result in increased milk yield. However, the ratio between RDS and RDP intakes in this study (36) was similar between barley and cottonseed meal and milo and brewers dried grain diets. The diets in this study delivered somewhat lower levels of postruminal starch than those of McCarthy et al. (78); they attributed the production response to increased passage and intestinal digestibility of starch, which increased glucose availability.

Oliveira et al. (99) fed lactating cows steam-rolled corn, dry-rolled sorghum, steam-flaked sorghum, and a mixture of the latter two sources. Dry matter intake was 4.6 kg higher for cows receiving the mixture compared with the steam-flaked sorghum. Intake of RDS ranged from 3.3 to 6 kg/d; corresponding postruminal starch deliveries ranged from .4 to 1.6 kg/d and total tract digestibilities from 71.6 to 90.8%. There were no differences in milk production, although milk protein percentage was highest for cows receiving the lowest level of postruminal starch. Poore et al. (107) fed cows dry-rolled or steam-flaked milo at similar DM and starch intakes. Steam flaking increased the amount of starch digestion in the rumen (4.7 vs. 3.3 kg) and decreased the amount delivered to the small intestine. Steam flaking increased total tract digestibility by 17 percentage units and increased milk yield by 3.4 kg. These four studies provide no consistent pattern that suggests that providing increased postruminal starch enhances milk yield or alters composition. In two of the studies (37, 107), increased ruminal starch digestion resulted in increased milk production. Three of the studies were confounded by differences in DM and energy intakes associated with site digestion. One common point in three of the four studies was that milk production was highest for diets that provided total tract digestibilities of starch of 90%. If the lower gut is provided with large

quantities (4.7 kg) of starch (78), increased dietary glucose may be available to support increased lactose synthesis. However, postruminal deliveries of .4 and 2.2 kg did not appear to influence performance. These data suggest that ruminal starch digestion plays a significant role in supplying energy-yielding substrates for milk synthesis, primarily through propionate.

To obtain a general perspective on the influence of site of starch digestion on production performance, separate correlations were conducted between production performance and intakes associated with different sites in the gastrointestinal tract. The database used was described previously. Dry matter intake was correlated positively to NE_L intake and to all starch intake measures (Table 10). These parameters also were correlated to milk and protein yields in a fashion similar to DMI. Milk protein percentage and fat yield or percentage were not related to starch intake or site digestibility parameters.

Regression analyses were conducted on the same dataset for two objectives: 1) to evaluate how energy parameters and those associated with site digestion would relate to milk yield and 2) to consider all nutrient parameters and effect of their digestion site on milk yield. In addition to carbohydrate parameters previously described, protein and fat intakes, ruminal, postruminal, and total tract digestibilities for protein were determined. Ruminal protein degradabilities, when not listed in the study, were estimated from Nocek and Russell (97). Intestinal and total tract digestibilities were derived from individual studies; when these were not listed, 65% intestinal digestibility was assumed. When values for ingredient fat were not listed for the total ration, they were obtained from NRC (89). The analysis used to derive regressions for energy and nutrient parameters was the Maximum R^2 Improvement Program (122) for the dependent variable milk.

When considering net energy of lactation alone, the resultant R^2 was .48 ($P < .001$, $Y = .679 \times X + 7.88$). The addition of rumen and IDS increased the R^2 to .58 (Table 11). Additional energy parameters did not further enhance the coefficient of determination. This would suggest that NSC digestion both ruminally and postruminally served to explain more of the variation associated with predicting milk yield than NE_L alone. Inclusion of ruminally

TABLE 9. Effect of starch on site and extent of starch digestion and production performance in Holstein cows.

Reference	Main carbohydrate source	DMI	Starch							Milk	FCM	Protein	
			Intake	Rumen degradation		APRD ¹		ATTD ²					
			(kg/d)	(% of intake)	(kg)	(% of intake)	(% of ruminal escape)	(kg/d)	(%)			(kg/d)	
(78)	GSC ³	23.8	10.6	5.2	49.1	4.7	44.3	87.0	93.2	35.6 ^a	28.4	3.18	1.13 ^a
	SRB ⁴	20.7	8.4	6.5	77.4	1.6	19.0	84.2	96.7	32.5 ^b	27.3	3.16	1.02 ^b
(37)	DRB and CSM ⁵	25.3	8.3 ^a	6.2 ^a	74.7	1.5	18.1	69.0	92.2	37.4 ^a	34.3 ^a	2.90	.99
	DRB and BDG ⁶	23.7	7.1 ^b	4.9 ^b	69.0	1.3	18.3	60.9	87.8	34.9 ^b	31.4 ^b	3.00	.94
	DRM and CSM ⁷	24.3	7.4 ^b	4.6 ^b	62.2	1.4	23.0	48.6	80.5	34.2 ^b	33.6 ^a	3.00	1.01
	DRM and BDG ⁸	23.8	5.8 ^c	2.8 ^c	48.3	1.9	32.7	62.3	80.5	34.6 ^b	34.8 ^a	2.80	.97
(99)	SRC ⁹	26.3 ^{ab}	6.73	4.4	65.0	1.2	17.8	51.7	83.0	30.2	31.0	3.01 ^b	.93
	DRM ¹⁰	26.1 ^{ab}	6.73	3.3	48.0	1.6	23.8	45.4	71.6	29.4	30.6	2.95 ^b	.90
	SFM ¹¹	24.8 ^b	7.11	6.0	85.0	.4	5.6	38.9	90.8	31.0	30.8	3.10 ^a	.95
	50% DRM and 50% SFM	29.4 ^a	8.32	5.6	67.0	1.4	16.8	49.5	83.3	30.9	29.7	3.06 ^{ab}	.91
(107)	DRM	20.7	6.7	3.3	49.2	2.2	32.8	62.8	80	28.1 ^b	28.2	2.90 ^a	.82
	SFM	20.3	6.4	4.7	73.9	1.6	25.0	91.8	97	31.5 ^a	29.8	2.98 ^b	.89

^{a,b}Means with different superscripts within study are different ($P < .05$).

¹Apparent postruminal digestion.

²Apparent total tract digestibility.

³Ground shelled corn.

⁴Steam-rolled barley.

⁵Dry-rolled barley and cottonseed meal.

⁶Dry-rolled barley and brewers dried grains.

⁷Dry-rolled milo and cottonseed meal.

⁸Dry-rolled milo and brewers dried grains.

⁹Steam-rolled corn.

¹⁰Dry-rolled milo.

¹¹Steam-flaked milo.

TABLE 10. Simple correlations between production performance and starch intakes at difference sites in the gastrointestinal tract.¹

Independent variable	DMI	NE _L	Starch intake	Starch digestibility			Total tract
				Rumen	Rumen escape	Postrumen	
	(kg/d)	(Mcal/d)	(kg/d)			(kg/d)	
DMI, kg	. . .	.97 ^b	.50 ^b	.43 ^b	.42 ^b	.38 ^a	.48 ^b
Milk, kg	.64 ^a	.70 ^b	.56 ^b	.40 ^b	.60 ^b	.53 ^b	.51
Protein, %	-.21	-.26	.03	.12	.13	.01	.10
Protein, kg	.62 ^a	.65 ^b	.62 ^b	.52 ^b	.59 ^b	.54 ^b	.60 ^b
Fat, %	-.20	-.17	-.09	-.02	-.05	-.03	-.02
Fat, kg	-.14	-.10	-.03	.03	-.11	.02	.03

^{a,b}Correlation coefficient differs significantly from zero, $P < .05$, $P < .01$.

¹References: (8, 9, 16, 36, 37, 59, 60, 78, 83, 99, 107, 117, 130, 134).

degradable NDF did not enhance the predictability of milk yield.

Separate energy and protein intakes and site digestibilities were considered in predicting milk yield. Seven independent variables entered the equation, which maximized the coefficient of determination (R^2). Intakes of NE_L, RDS, and IDS did not enter the equation as significant predictors. However, starch intake and total tract starch digestibility were included. Several protein variables entered the equation, demonstrating that protein plays a key role in predicting milk yield in relation to total starch intake and digestibility.

The production database was used to determine whether an optimum combination of RDS and IDS level would maximize milk yield. A matrix of 60 diets was formulated, ranging in RDS intake from 1.8 to 6.4 kg/d and in IDS intake from .5 to 2.4 kg/d. Protein intake and digestibility parameters, NE_L and fat intake, were kept constant. Equation [1] (Table 11) was used to predict milk yield. Figure 13 illustrates the surface response of milk yield as influenced by RDS and IDS at equal energy intakes adequate to support 32 kg of milk (89). This response indicates that maximum milk production was associated with the greatest intake of both RDS and IDS. At low IDS, increasing level of RDS over a wide range had a relatively small impact on milk yield compared with increasing IDS at any level of RDS. Although the relationship suggests that starch digested in the small intestine is utilized more efficiently for milk synthesis than starch in the rumen, the magnitude of this

difference is questionable. In beef cattle, Owens et al. (101) concluded that RDS was used for gain 70% as efficiently as IDS.

CONCLUSIONS

Site of starch digestion and digestion within site can be manipulated. Processing and feeding management strategies largely influence site digestion.

There appear to be no limitations to intestinal starch digestion, because postruminal starch digestion increases as more starch escapes the rumen. Efficiency of postruminal

TABLE 11. Equations developed to maximize R^2 for the dependent variable, milk yield, considering energy intake, nutrient intakes, and site digestion variables.¹

Equation	Variable	Coefficient	P
[1]	Intercept	6.07	
	NE _L	.55	.0001
	Rumen degradable starch	.48	.206
	Intestinally digestible starch	2.28	.009
	R ²	.58	
[2]	Intercept	5.08	
	Starch intake	-3.29	.003
	Total tract starch digestibility	5.29	.001
	Fat intake	1.31	.165
	Rumen degradable protein	206.85	.008
	Rumen escape protein	2.72	.003
	Intestinal digestible protein	211.62	.007
	Total tract protein digestibility	-202.96	.009
	R ²	.90	

¹References used to develop relationships: (8, 9, 16, 36, 37, 59, 60, 78, 83, 99, 107, 117, 130, 134).

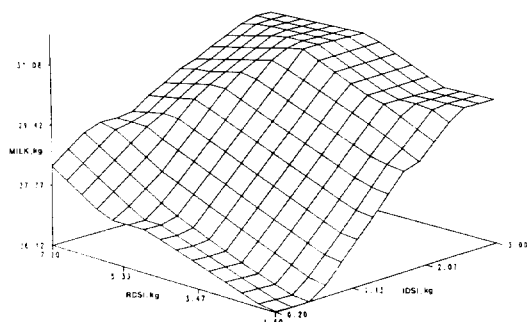


Figure 13. Predicted milk yield as a result of varying rumen degradable starch intake [(RDSI), kg/d] and intestinally digestible starch intake [(IDSI), kg/d] at equal NE_L intakes for dairy cattle. Milk yield (kg/d) = $6.07 + .55 \times (NE_L) + .48 \times (RDSI) + 2.28 \times (IDSI)$.

starch digestion, however, decreases with increased starch passage into the small intestine.

With few exceptions, there is no net glucose absorption at the portal vein, and plasma glucose levels remain relatively unaffected as digestible dietary starch is presented to the intestine. There appears to be a high demand for glucose for oxidative metabolism at the visceral tissue level. Exogenous glucose supply may spare endogenously synthesized glucose for gut metabolism, allowing more to be directed to the mammary gland.

Production studies yield no clear evidence that postruminal starch digestion enhances milk yield or changes composition. However, there is evidence to suggest that postruminally digested starch may be used more efficiently for milk synthesis than starch digested in the rumen.

REFERENCES

- 1 Aman, P., and K. Hessleman. 1985. A enzymatic method for analysis of total mixed-linkage β -glucans in cereal grains. *J. Cereal Sci.* 3:231.
- 2 Acuirre, E. O., A. L. Goetch, and F. N. Owens. 1984. Corn grain processing and site of digestion by heifers. Page 190 in *Anim. Sci. Res. Rep. MP116*, Oklahoma State Univ., Stillwater.
- 3 Bailey, R. W., and M. J. Ulyatt. 1970. Pasture quality and ruminant nutrition II. Carbohydrate and lignin composition of detergent extracted residues from pasture grasses and legumes. *N.Z. J. Agric. Res.* 13: 591.
- 4 Baird, G. B., J. G. Van der Walt, and E. N. Bergman. 1983. Whole-body metabolism of glucose and lactate in productive sheep and cows. *Br. J. Nutr.* 50:249.
- 5 Baldwin, R. L., J. France, D. E. Beever, M. Gill, and J.H.M. Thornley. 1987. Metabolism of the lactating cow. III. Properties of mechanistic models suitable for evaluation of energetic relationships and factors involved in the partition of nutrients. *J. Dairy Res.* 54:133.
- 6 Brown, W. H., J. W. Stull, F. O. Dafaala, and P. W. Riley. 1966. Comparison of milo and barley for lactating cows. *J. Dairy Sci.* 49:386.
- 7 Campling, R. C., and M. Freer. 1962. The effect of specific gravity on the mean time of retention of inert particles in the alimentary tract of the cow. *Br. J. Nutr.* 20:29.
- 8 Casper, D. P., and D. J. Schingoethe. 1989. Lactational response of dairy cows to diets varying in ruminal solubilities of carbohydrates and crude protein. *J. Dairy Sci.* 72:928.
- 9 Casper, D. P., D. J. Schingoethe, and W. A. Eisenbeisz. 1990. Response of early lactating dairy cows fed diets varying in source of nonstructural carbohydrate and crude protein. *J. Dairy Sci.* 73:1039.
- 10 Clary, J. J., G. E. Mitchell, Jr., C. O. Little, and N. W. Bradley. 1969. Pancreatic amylase activity from ruminants fed different rations. *Can. J. Physiol. Pharmacol.* 47:161.
- 11 Cole, N. A., R. R. Johnson, and F. N. Owens. 1976. Influence of roughage level and corn processing method on the site and extent of digestion by beef steers. *J. Anim. Sci.* 43:490.
- 12 Conrad, H. R. 1966. Symposium on factors influencing the voluntary intake of herbage by ruminants: physiological and physical factors limiting feed intake. *J. Anim. Sci.* 25:227.
- 13 Coombe, V. B., and R. C. Siddons. 1973. Carbohydrates of the bovine small intestine. *Br. J. Nutr.* 30: 269.
- 14 Crawford, R. J., Jr., B. J. Shriver, G. A. Varga, and W. H. Hoover. 1983. Buffer requirements for maintenance of pH during fermentation of individual feeds in continuous cultures. *J. Dairy Sci.* 66:1881.
- 15 Demeyer, A., and C. J. VanNevel. 1979. Effect of defaunation on the metabolism of rumen microorganisms. *Br. J. Nutr.* 42:515.
- 16 DePeters, E. J., and S. J. Taylor. 1985. Effects of feeding corn or barley on composition of milk and diet digestibility. *J. Dairy Sci.* 68:2027.
- 17 deVisser, H., and A.A.M. deGroot. 1980. The influence of the starch and sugar content of concentrates on feed intake, rumen fluid, production and composition of milk. Page 2 in *Proc. 4th Int. Conf. Prod. Dis. Farm Anim.*, Munchen, Germany.
- 18 deVisser, H., P. L. Van der Togt, and S. Tamminga. 1990. Structural and nonstructural carbohydrates in concentrate supplements to silage based diets. 1. Feed intake and milk production. *Neth. J. Agric. Sci.* 38:487.
- 19 Dixon, R. M., and L. P. Milligan. 1985. Removal of digesta components from the rumen of steers determined by sieving techniques and fluid particulate and microbial markers. *Br. J. Nutr.* 53:347.
- 20 Durkwa, L. 1983. Length and specific gravity of particles passed from the rumen and changes in ingesta specific gravity. Ph.D. Diss., Univ. Vermont, Burlington.

- 21 Ehle, F. R. 1984. Influence of feed particle density on particulate passage from rumen of Holstein cow. *J. Dairy Sci.* 67:693.
- 22 Ehle, F. R., M. R. Murphy, and J. H. Clark. 1982. In situ particle size reduction and the effect of particle size on degradation of crude protein and dry matter in the rumen of dairy steers. *J. Dairy Sci.* 65:963.
- 23 Ewing, D. L., and D. E. Johnson. 1987. Corn particle starch digestion, passage and size reduction in beef steers: a dynamic model. *J. Anim. Sci.* 64:1194.
- 24 Ewing, D. L., D. E. Johnson, and W. V. Rimpler. 1986. Corn particle passage and size reduction in the rumen of beef steers. *J. Anim. Sci.* 63:1509.
- 25 Fluharty, F. L., and S. C. Loerch. 1989. Chemical treatment of ground corn to limit ruminal starch digestion. *Can. J. Anim. Sci.* 69:173.
- 26 Fox, D. G., C. J. Sniffen, J. D. O'Connor, J. B. Russell, and P. J. Van Soest. 1988. The Cornell net carbohydrate and protein system for evaluating cattle diets. Cornell Univ. Agric. Exp. Stn. Bull., Ithaca, NY.
- 27 French, D. 1973. Chemical and physical properties of starch. *J. Anim. Sci.* 37:1048.
- 28 Fuskiki, T., and K. Iwai. 1989. Two hypotheses on the feedback regulation of pancreatic enzyme secretion. *Fed. Am. Soc. Exp. Biol. J.* 3:121.
- 29 Galyean, M. L., D. G. Wagner, and R. R. Johnson. 1976. Site and extent of starch digestion in steers fed processed corn rations. *J. Anim. Sci.* 43:1088.
- 30 Galyean, M. L., D. G. Wagner, and F. N. Owens. 1979. Corn particle size and site and extent of digestion by steers. *J. Anim. Sci.* 49:204.
- 31 Galyean, M. L., D. G. Wagner, and F. N. Owens. 1981. Dry matter and starch disappearance of corn and sorghum as influenced by particle size and processing. *J. Dairy Sci.* 64:1804.
- 32 Garcia, J. A., C. B. Theurer, and W. H. Hale. 1981. Digestao ruminal e pos-ruminal do amido, em novilhos alimentados com dois niveis de grau de sorgo floculado. *Rev. Soc. Bras. Zootec.* 10:235.
- 33 Gross, K. L., D. L. Harmon, J. E. Minton, and T. B. Avery. 1989. Effect of isoenergetic infusions of propionate or glucose on portal-drained visceral nutrient flux and concentrations of hormones in lambs maintained by total intragastric infusion. *J. Anim. Sci.* 67(Suppl. 1):554.(Abstr.)
- 34 Hagenmeister, H. W., Kaufmann, and E. Pfeffer. 1976. Factors influencing the supply of nitrogen and amino acids to the intestine of dairy cows. Page 425 in *Protein metabolism and nutrition*. D.J.A. Cole, K. N. Boorman, P. J. Buttery, D. Lewis, R. J. Neale, and H. Swan, ed. Butterworths, Inc., Boston, MA.
- 35 Hale, W. H. 1973. Influence of processing on the utilization of grains (starch) by ruminants. *J. Anim. Sci.* 37:1075.
- 36 Herrera-Saldana, R., R. Gomez-Alarcon, M. Torabi, and J. T. Huber. 1990. Influence of synchronizing protein and starch degradation in the rumen on nutrient utilization and microbial protein synthesis. *J. Dairy Sci.* 73:142.
- 37 Herrera-Saldana, R., and J. T. Huber. 1989. Influence of varying protein and starch degradabilities on performance of lactating cows. *J. Dairy Sci.* 72:1477.
- 38 Hespell, R. E., and M. P. Bryant. 1979. Efficiency of rumen microbial growth: influence of some theoretical and experimental factors on Y(ATP). *J. Anim. Sci.* 49:1640.
- 39 Hibberd, C. A., D. G. Wagner, R. L. Hintz, and D. D. Griffin. 1983. Effect of sorghum grain variety and processing method on site and extent of starch digestion in steers. Page 28 in *Anim. Sci. Res. Rep. MP114*, Oklahoma State Univ., Stillwater.
- 40 Hinman, D. D., and R. R. Johnson. 1974. Influence of processing methods on digestion of sorghum starch in high concentrate beef cattle rations. *J. Anim. Sci.* 49:417.
- 41 Hinman, D. D., and R. R. Johnson. 1974. Influence of degree of micronization on the site and extent of sorghum starch digestion in beef cattle fed high concentrate rations. *J. Anim. Sci.* 39:958.
- 42 Hino, T., and J. B. Russell. 1985. The effect of reducing equivalent disposal and NADH/NAD on the deamination of amino acids by intact and cell-free extracts of rumen microorganism. *Appl. Environ. Microbiol.* 50:1368.
- 43 Hino, T., and J. B. Russell. 1986. Relative contribution of ruminal bacteria and protozoa to the degradation of protein in vitro. *J. Anim. Sci.* 64:261.
- 44 Holmes, J.H.G., M. J. Drennan, and W. N. Garrett. 1970. Digestion of steam-processed milo by ruminants. *J. Anim. Sci.* 31:409.
- 45 Huber, J. T., N. L. Jacobson, A. D. McGilliard, and R. S. Allen. 1961. Utilization of carbohydrates introduced directly into the omaso-abomasal area of the stomach of cattle of various ages. *J. Dairy Sci.* 44:321.
- 46 Huntington, G. B. 1982. Portal blood flow and net absorption of ammonia-nitrogen, urea nitrogen, and glucose in nonlactating Holstein cows. *J. Dairy Sci.* 65:1155.
- 47 Huntington, G. B. 1984. Net absorption of glucose and nitrogenous compounds by lactating Holstein cows. *J. Dairy Sci.* 67:1919.
- 48 Huntington, G. B., and P. J. Reynolds. 1986. Net absorption of glucose L-lactate, volatile fatty acids, and nitrogenous compounds by bovine given abomasal infusions of starch or glucose. *J. Dairy Sci.* 69:2428.
- 49 Huntington, G. B., P. J. Reynolds, and H. F. Tyrrell. 1983. Net absorption and ruminal concentrations of metabolites in nonpregnant dry Holstein cows before and after intraruminal acetic acid infusion. *J. Dairy Sci.* 66:1901.
- 50 Hvelplund, T., and J. Madsen. 1985. Amino acid passage to the small intestine in dairy cows compared with estimates of microbial protein and undegraded dietary protein from analysis on the feed. *Acta Agric. Scand.* 25(Suppl.):21.
- 51 Institut National de la Recherche Agronomique. 1989. Ruminant nutrition: recommended allowances and feed tables. R. Jarrige, ed. John Libbey Eurotext, Montrouge, France.
- 52 Isaacson, J. R., F. C. Hinds, M. P. Bryant, and F. N. Owens. 1975. Efficiency of energy utilization by mixed rumen bacteria in continuous culture. *J. Dairy Sci.* 58:1645.
- 53 Istasse, L., and E. R. Orskov. 1984. The effects of intermittent and continuous infusions of propionic

- acid on plasma insulin. *Can. J. Anim. Sci.* 64(Suppl.):148.
- 54 Janes, A. N., T.E.C. Weekes, and D. G. Armstrong. 1985. Carbohydrase activity in the pancreatic tissue and small intestine mucosa of sheep fed dried-grass or ground maize-based diets. *J. Agric. Sci. (Camb.)* 104:435.
 - 55 Janes, A. N., T.E.C. Weekes, and D. G. Armstrong. 1985. Absorption and metabolism of glucose by the mesenteric-drained viscera of sheep fed on dried-grass or ground, maize-based diets. *Br. J. Nutr.* 54: 449.
 - 56 Journet, M., and R. Verite. 1979. Predicting equations of duodenal N flow in dairy cattle. Effects of level of feeding and proportion of concentrate in the diet. *Ann. Rech. Vet.* 10:303.
 - 57 Kartchner, R. J. 1972. Starch digestion in the bovine as influenced by level and processing of sorghum grain. M.S. Thesis, Univ. Arizona, Tucson.
 - 58 Katoh, K., and T. Yajima. 1989. Effects of butyric acid and analogues on amylase release from pancreatic segments of sheep and goats. *Eur. J. Physiol.* 413:256.
 - 59 Klusmeyer, T. H., G. L. Lynch, and J. H. Clark. 1991. Effects of calcium salts of fatty acids and proportion of forage in diet on ruminal fermentation and nutrient flow to duodenum of cows. *J. Dairy Sci.* 74:2220.
 - 60 Klusmeyer, T. H., R. D. McCarthy, Jr., J. H. Clark, and D. R. Nelson. 1990. Effects of source and amount of protein on ruminal fermentation and passage of nutrients to the small intestine of lactating cows. *J. Dairy Sci.* 73:3526.
 - 61 Kreikemeier, K. K., D. L. Harmon, R. T. Brandt, Jr., T. B. Avery, and D. E. Johnson. 1991. Small intestinal starch digestion in steers: effect of various levels of abomasal glucose, corn starch and corn dextrin infusion on small intestinal disappearance and net glucose absorption. *J. Anim. Sci.* 69:328.
 - 62 Kreikemeier, K. K., D. L. Harmon, J. P. Peters, K. L. Gross, C. K. Armendariz, and C. R. Krehbiel. 1990. Influence of dietary forage and feed intake on carbohydrase activities and small intestinal morphology in calves. *J. Anim. Sci.* 68:2916.
 - 63 Kronfeld, D. S. 1976. The potential importance of the proportions of glucogenic, lipogenic and aminogenic nutrients in regard to the health and productivity of dairy cows. *Adv. Anim. Physiol. Anim. Nutr.* 7:5.
 - 64 Kronfeld, D. S., G. P. Mayer, J. M. Robertson, and F. Raggi. 1963. Depression of milk secretion during insulin administration. *J. Dairy Sci.* 46:553.
 - 65 Kronfeld, D. S., F. Raggi, and C. F. Ramberg. 1968. Mammary blood flow and ketone metabolism in normal, fasted and ketotic cows. *Am. J. Physiol.* 215: 218.
 - 66 Larsen, H. J., G. E. Stoddard, N. L. Jacobson, and R. S. Allen. 1956. Digestion and absorption of various carbohydrates posterior to the ruminoreticular area of the young bovine. *J. Anim. Sci.* 15:473.
 - 67 Lee, R. W., M. L. Galyean, and G. P. Lofgreen. 1982. Effects of mixing whole shelled and steam-flaked corn in finishing diets on feedlot performance and site and extent of digestion in beef steers. *J. Anim. Sci.* 55:475.
 - 68 Lomax, M. A., and D. G. Baird. 1983. Blood flow and nutrient exchange across the liver and gut of the dairy cow. *Br. J. Nutr.* 49:481.
 - 69 MacGregor, C. A. 1979. Factors affecting the utilization of protein and carbohydrate in lactating dairy cattle. Ph.D. Diss., Univ. Maine, Orono.
 - 70 MacGregor, C. A., M. R. Strokes, W. H. Hoover, H. A. Leonard, L. L. Junkins, Jr., C. J. Sniffen, and R. W. Mailman. 1983. Effect of dietary concentration of total nonstructural carbohydrate on energy and nitrogen metabolism and milk production of dairy cows. *J. Dairy Sci.* 66:39.
 - 71 MacRae, J. C., P. J. Buttery, and D. E. Beever. 1988. Nutrient interaction in the dairy cow. Page 55 in *Nutrition and lactation in the dairy cow*. P. G. Garnsworthy, ed. Butterworths, London, Engl.
 - 72 Malenstein, A., A. Th. Van't Klooster, and J. W. Corne. 1988. Degradability of various types of starch by incubation with rumen fluid or with bacterial amylases. *J. Anim. Physiol. Anim. Nutr.* 59:225.
 - 73 Mandell, I. B., H. H. Nicholson, and G. I. Christian. 1988. The effects of barley processing on nutrient digestion within the gastrointestinal tract of beef cattle fed mixed diets. *Can. J. Anim. Sci.* 68:191.
 - 74 Martin, P. A., and P. C. Thomas. 1988. Dietary manipulation of the yield and composition of milk: effect of dietary inclusions of barley and oats in untreated and formaldehyde-treated forms on milk fatty acid composition. *J. Sci. Food Agric.* 43:145.
 - 75 Mayes, R. W., and E. R. Ørskov. 1974. The utilization of gelled maize starch in the small intestine of sheep. *Br. J. Nutr.* 32:143.
 - 76 McBride, B. W., and L. P. Milligan. 1985. Influence of feed intake and starvation on the magnitude of Na^+ , K^+ , -ATPase (E.C. 3.6.1.3)—dependent respiration in duodenal mucosa of sheep. *Br. J. Nutr.* 53: 605.
 - 77 McBurney, M. I., P. J. Van Soest, and L. E. Chase. 1981. Cation exchange capacity of various feedstuffs in ruminant rations. Page 16 in *Proc. Cornell Nutr. Conf. Feed Manuf.*, Syracuse, NY.
 - 78 McCarthy, R. D., Jr., T. H. Klusmeyer, J. L. Vicini, J. H. Clark, and D. R. Nelson. 1989. Effect of source of protein and carbohydrate on ruminal fermentation and passage of nutrients to the small intestine of lactating cows. *J. Dairy Sci.* 72:2002.
 - 79 McNeil, J. W., G. D. Potter, and J. K. Riggs. 1971. Ruminal and post-ruminal carbohydrate utilization in steers fed processed sorghum grain. *J. Anim. Sci.* 33: 1371.
 - 80 Mertens, D. R. 1983. Using neutral detergent fiber to formulate dairy rations and estimate the energy content of forages. Page 60 in *Proc. Cornell Nutr. Conf. Feed Manuf.*, Syracuse, NY.
 - 81 Mertens, D. R. 1985. Recent concepts useful in optimizing nutrition of dairy cows. Page 99 in *Mon-santo Tech. Symp.*, Bloomington, MN.
 - 82 Mertens, D. R. 1988. Balancing carbohydrates in dairy rations. Page 150 in *Large Dairy Herd Manage. Conf.* March 16-18, Syracuse, NY.
 - 83 Metwally, H. M. 1989. Effects of dietary carbohydrate and protein sources on ruminal fermentation and protein supply to the small intestine. Ph.D. Diss., Univ. Minnesota, St. Paul.
 - 84 Moe, P. W., and H. F. Tyrrell. 1976. Estimating metabolizable and net energy of feeds. *Proc. 1st Int. Symp. Feed Composition Anim. Nutrient Requirements Computerization Diets.* Int. Feedstuffs Inst.,

- Logan, UT.
- 85 Moran, J. B. 1986. Cereal grains in complete diets for dairy cows: a comparison of rolled barley, wheat and oats and of three methods of processing oats. *Anim. Prod.* 43:217.
 - 86 Morgan, D. A., H. Ryan, and P. K. Upton. 1989. Cereal digestion in the ruminant. 1. Nutrient passage from the rumen and glucose kinetics in sheep fed unground barley in unground maize diets. *Ir. J. Agric. Res.* 28:35.
 - 87 Moseley, G., and J. R. Jones. 1984. The physical digestion of perennial ryegrass (*Lolium perenne*) and white clover (*Trifolium repens*) in the foregut of sheep. *Br. J. Nutr.* 52:381.
 - 88 Murphy, M. R., R. L. Baldwin, and L. J. Koong. 1982. Estimation of stoichiometric parameters for rumen fermentation of roughage and concentrate diets. *J. Anim. Sci.* 55:411.
 - 89 National Research Council. 1988. Nutrient requirements of dairy cattle. 6th rev. ed. Natl. Acad. Press, Washington, DC.
 - 90 Nicoletti, J. M., C. L. Davis, R. B. Hespell, and J.A.Z. Leedle. 1985. Enumeration and presumptive identification of bacteria from the small intestine of sheep. *J. Dairy Sci.* 67:1227.
 - 91 Nocek, J. E., inventor. 1986. Novel method for quantitating structural and nonstructural carbohydrates in feedstuffs. US Pat. No. 4,617,276.
 - 92 Nocek, J. E., 1987. Characterization of in situ dry matter and nitrogen of various corn grain forms. *J. Dairy Sci.* 70:2291.
 - 93 Nocek, J. E. 1988. In situ and other methods to estimate ruminal protein and energy digestibility: a review. *J. Dairy Sci.* 71:2051.
 - 94 Nocek, J. E., D. G. Braund, R. L. Steele, and C. A. MacGregor, inventors. 1986. Method of formulating dairy cow rations based on carbohydrate regulation. US Pat. No. 4,615,891.
 - 95 Nocek, J. E., and R. A. Kohn. 1987. Initial particle form and size on change in functional specific gravity of alfalfa and timothy hay. *J. Dairy Sci.* 70:1850.
 - 96 Nocek, J. E., and R. A. Kohn. 1988. In situ particle size reduction of alfalfa and timothy hay as influenced by form and particle size. *J. Dairy Sci.* 71:932.
 - 97 Nocek, J. E., and J. B. Russell. 1988. Protein and energy as an integrated system. Relationship of ruminal protein and carbohydrates availability to microbial synthesis and milk production. *J. Dairy Sci.* 71:2070.
 - 98 Oke, B. O., and S. C. Loerch. 1989. The effects of dietary levels and formaldehyde treatment of corn on-site and extent of starch digestion in sheep. *J. Anim. Sci.* (Suppl. 1):67:538.(Abstr.)
 - 99 Oliveira, J., J. T. Huber, D. Ben-Ghedalia, and M. Pessarakli. 1990. Effect of sorghum grain processing on the performance of lactating dairy cows. *J. Dairy Sci.* (Suppl. 1):73:127.(Abstr.)
 - 100 Ørskov, E. R. 1986. Starch digestion and utilization in ruminants. *J. Anim. Sci.* 63:1624.
 - 101 Owens, F. N., R. A. Zinn, and Y. K. Kim. 1986. Limits to starch digestion in the ruminants small intestine. *J. Anim. Sci.* 63:1634.
 - 102 Pehrson, B., U. Johnson, and M. Knutson. 1981. The digestion of starch in the small intestine of dairy cows. *Zentralbl. Veterinaarmed. Reihe A* 28:241.
 - 103 Pehrson, B., and M. Knutson. 1980. Glucose and lactose absorption from the small intestine of dairy cows. *Zentralbl. Veterinaarmed. Reihe A* 27:644.
 - 104 Pethick, D. W., D. B. Lindsay, P. J. Barker, and R. T. Northrup. 1981. Acetate supply and utilization by tissues of sheep in vivo. *Br. J. Nutr.* 46:97.
 - 105 Poppi, D. P., D. J. Minson, and J. H. Ternouth. 1981. Studies of cattle and sheep eating leaf and stem fractions of grass III. The retention time in the rumen of large feed particles. *Aust. J. Agric. Res.* 32:123.
 - 106 Poppi, D. P., B. W. Norton, D. J. Minson, and R. E. Hendrickson. 1980. The validity of the critical size theory for particles leaving the rumen. *J. Agric. Sci. (Camb.)* 94:275.
 - 107 Poore, M. H., J. A. Moore, R. S. Swingle, W. H. Brown, and F. M. Whiting. 1989. Influence of forage quality and sorghum grain processing in diets formulated to contain 25% forage NDF on milk production by Holstein cows. *J. Dairy Sci.* (Suppl. 1):72:480.(Abstr.)
 - 108 Remillard, R. L., and D. E. Johnson. 1984. Starch digestion in the bovine small intestine with and without buffer amylase infusion. *J. Anim. Sci.* (Suppl. 1):59:440.(Abstr.)
 - 109 Reynolds, P. J., and G. B. Huntington. 1988. Net absorption of organic acids by lactating Holstein cows. *J. Dairy Sci.* 71:124.
 - 110 Reynolds, C. K., and G. B. Huntington. 1988. Partition of portal-drained visceral net flux in beef steers 1. Blood flow and net flux of oxygen, glucose and nitrogenous compounds across stomach and post-stomach tissues. *Br. J. Nutr.* 60:539.
 - 111 Reynolds, C. K., and G. B. Huntington. 1988. Partition of portal-drained visceral net flux in beef steers. 2. Net flux of volatile fatty acids, D-B-hydroxy butyrate and L-lactate across stomach and post-stomach tissues. *Br. J. Nutr.* 60:553.
 - 112 Reynolds, C. K., G. B. Huntington, H. F. Tyrrell, and P. J. Reynolds. 1988. Net portal-drained visceral and hepatic metabolism of glucose, L-Lactate, and nitrogenous compounds in lactating Holstein cows. *J. Dairy Sci.* 71:1803.
 - 113 Reynolds, C. K., H. F. Tyrrell, and P. J. Reynolds. 1991. Effects of diet forage-to-concentrate ratio and intake on energy metabolism in growing beef heifers: whole body energy and nitrogen balance and visceral heat production. *J. Nutr.* 121:994.
 - 114 Reynolds, C. K., H. F. Tyrrell, and P. J. Reynolds. 1991. Effects of diet forage-to-concentrate ratio and intake on energy metabolism in growing beef heifers: net nutrient metabolism by visceral tissues. *J. Nutr.* 121:1004.
 - 115 Rich-Denson, C., and R. E. Kimura. 1988. Evidence in vivo that most of the intraluminally absorbed glucose is absorbed intact into the portal vein and not metabolized to lactate. *Biochem. J.* 254:931.
 - 116 Robinson, P. H., and J. J. Kennelly. 1988. Influence of ammoniation of high moisture barley on in situ rumen degradation and influence on rumen fermentation in dairy cows. *Can. J. Anim. Sci.* 68:839.
 - 117 Robinson, P. H., and J. J. Kennelly. 1989. Influence of high moisture barley on digestibility, kinetics of

- rumen ingesta turnover, and milk production in dairy cows. *Can. J. Anim. Sci.* 69:195.
- 118 Robinson, P. H., and J. J. Kennelly. 1990. Evaluation of a duodenal cannula for dairy cattle. *J. Dairy Sci.* 73:3146.
 - 119 Rooney, L. W., and R. L. Pfugfelder. 1986. Factors affecting starch digestibility with special emphasis on sorghum and corn. *J. Anim. Sci.* 63:1607.
 - 120 Russell, J. R., A. W. Young, and N. A. Jorgensen. 1981. Effect of dietary corn starch intake on ruminal small intestinal and large intestinal starch digestion in cattle. *J. Anim. Sci.* 52:1170.
 - 121 Rust, S. R., F. N. Owens, D. R. Gill, and R. W. Fent. 1979. Corn moisture, protein concentration and rumen digestion by feedlot steers. Page 143 in *Oklahoma Agric. Exp. Stn. MP-107*, Stillwater.
 - 122 SAS® User's Guide: Statistics, Version 5 Edition. 1985. SAS Inst., Inc., Cary, NC.
 - 123 Scharrer, E. 1976. Development of changes of sugar transport in the ovine small intestine. *Pfluegers Arch. Eur. J. Physiol.* 336:147.
 - 124 Schuldt, A. 1989. Einfluss der Getreideart, der technischen Behandlung von Getreide sowie der Ration-zusammensetzung auf Ort und Ausmass der Verdauung der Getreidestärke in Milchkuhen. Ph.D. Diss. Christian Albrechts Univ., Kiel, Germany.
 - 125 Siddons, R. C. 1968. Carbohydrase activities in the bovine digestive tract. *Biochem. J.* 108:839.
 - 126 Smith, D. 1969. Removing and analyzing nonstructural carbohydrates from plant tissues. *Wisconsin Agric. Exp. Stn. Res. Rep.* 41, Madison.
 - 127 Smith, L. W., B. T. Weinland, D. R. Waldo, and E. C. Leffel. 1983. Rate of plant cell wall particle size reduction in the rumen. *J. Dairy Sci.* 66:2124.
 - 128 Sniffen, C. J., and P. H. Robinson. 1984. Nutritional strategy. *Can. J. Anim. Sci.* 64:529.
 - 129 Spicer, L. A., C. B. Theurer, J. Souse, and T. H. Noon. 1986. Ruminal and post-ruminal utilization of nitrogen and starch from sorghum grain, corn, and barley based diets by beef steers. *J. Anim. Sci.* 62:521.
 - 130 Stokes, S. R. 1990. The effects of different carbohydrate and protein types on ruminal fermentation and microbial metabolism. Ph.D. Diss., West Virginia Univ., Morgantown.
 - 131 Sutton, J. D. 1979. Carbohydrate fermentation in the rumen—variations on the theme. *Proc. Nutr. Soc.* 38:275.
 - 132 Sutton, J. D. 1985. Digestion and absorption of energy substrates in the lactating cow. *J. Dairy Sci.* 68:3376.
 - 133 Sutton, J. D., J. D. Oldham, and I. C. Hart. 1980. Products of digestion, hormones and energy utilization in milking cows given concentrates containing varying proportions of barley or maize. Page 303 in *Energy Metabolism*. L. E. Mount, ed. Butterworths, London, Engl.
 - 134 Swingle, R. S., J. Moore, M. Poore, and T. Eck. 1990. Utilization of starch from processed sorghum grain. Page 52 in *Proc. Southwestern Nutr. Manage. Conf.* Feb 1-2, Tempe, AZ.
 - 135 Tamminga, S., P. Van der Togt, C. J. Van der Koelen, C. Meliefste, M. Luttikuis, and G.D.H. Claassen. 1989. The behavior of starch in the rumen of dairy cows. *Mededeling No. 14. Inst. Livest. Feeding Nutr.* Lelystad, Neth.
 - 136 Tamminga, S., A. M. Van Vuuren, C. J. Van der Koelen, R. S. Ketelaar, and P. L. Van der Togt. 1990. Ruminal behavior of structural carbohydrates, non-structural carbohydrates and crude protein from concentrate ingredients in dairy cows. *Neth. J. Agric. Sci.* 38:513.
 - 137 Theurer, C. B. 1986. Grain processing effects on starch utilization by ruminants. *J. Anim. Sci.* 63:1649.
 - 138 Thorne, M. J., L. U. Thompson, and D.J.A. Jenkins. 1983. Factors affecting starch digestibility and the glycemic response with special reference to legumes. *Am. Clin. J. Nutr.* 38:481.
 - 139 Tommervik, R. S., and D. E. Walden. 1969. Comparative feeding value of wheat, corn, barley, milo, oats and a mixed concentrate ration for lactating cows. *J. Dairy Sci.* 52:68.
 - 140 Turgeon, O. A., Jr., D. R. Brink, and R. A. Britton. 1983. Corn particle size mixtures, roughage level and starch utilization in finishing steer diets. *J. Anim. Sci.* 57:739.
 - 141 Van der Walt, J. G., G. D. Baird, and E. N. Bergman. 1983. Tissue glucose and lactate metabolism and interconversions in pregnant and lactating sheep. *Br. J. Nutr.* 50:267.
 - 142 VanHellen, R. W., T. A. Wilson, G. E. Mitchell, Jr., G. T. Schelling, and R. E. Tucker. 1978. Bovine amylase, insulin and glucose response to high and low concentrate diets. *J. Anim. Sci.* 47(Suppl. 1):445.(Abstr.)
 - 143 Van Soest, P. J. 1982. Nutritional ecology of the ruminant. O & B Books, Inc., Corvallis, OR.
 - 144 Van Soest, P.J. 1986. Soluble carbohydrates. Page 73 in *Proc. Cornell Nutr. Conf. Feed Manuf.*, Syracuse, NY.
 - 145 Veerasilerp, T. 1986. Site and extent of maize starch digestion in relation to stage of maize maturity in lactating cows. Ph.D. Diss., Christian Albrecht Univ., Kiel, Germany.
 - 146 Veira, A. M., and G. K. Macleod. 1980. Effects of physical form of corn and urea supplementation on the performance of male Holstein calves. *Can. J. Anim. Sci.* 60:931.
 - 147 Voight, J., B. Piatkowski, and R. Krawielitzki. 1978. Die Wirkung der Reihenfolge von Grobfutter und Konzentraten in der Fütterung auf die Kohlenhydratverdauung und die bakterielle Proteinsynthese im Pansen der Milchkuh. *Archiv. Tierernähr.* 28:67.
 - 148 Voight, J., B. Piatkowski, R. Krawielitzki, and K. O. Trautmann. 1977. Die Wirkung der physikalischen Form von Trockengrünfutter und verschiedener Starkequellen auf den Ort der Verdauung der Kohlenhydrate und des Proteins sowie die bakterielle Proteinsynthese im Pansen der Kuh. *Archiv. Tierernähr.* 27:393.
 - 149 Voight, J., B. Piatkowski, S. Nagel, and R. Krawielitzki. 1976. Einfluss der physikalischen Form von Weizenstroh und verschiedener Starkequellen auf den Ort der Verdauung von starke und Rohzellulose bei der Kuh. *Archiv. Tierernähr.* 26:345.
 - 150 Waldo, D. 1973. Extent and partition of cereal grain starch digestion in ruminants. *J. Anim. Sci.* 37:1062.
 - 151 Watson, M. J., G. P. Savage, I. Brown, and D. G. Armstrong. 1972. Sites of disappearance of apparently digestible cellulose and apparently digestible

- linked glucose polymers in the digestive tract of a cow receiving dried grass-concentrate diets. *Proc. Nutr. Soc.* 31:991.
- 152 Webster, A.J.F., R. D. Dewhurst, and C. J. Waters, 1988. Alternative approaches to the characterization of feedstuffs for ruminants. Page 167 in *Recent advances in animal nutrition*. W. Haresign and D.J.A. Cole, ed. Butterworths, London, Engl.
- 153 Weighart, M., R. Sleptis, J. M. Elliot, and D. F. Smith. 1986. Glucose absorption and hepatic gluconeogenesis in dairy cows fed diets varying in forage content. *J. Nutr.* 116:839.
- 154 Weiss, W. P., G. R. Fisher, and G. M. Eriksson. 1989. Effect of source of neutral detergent fiber and starch on nutrient utilization by dairy cows. *J. Dairy Sci.* 72:2308.
- 155 White, R. G., V. J. Williams, and R.J.H. Morris. 1971. Acute in vivo studies on glucose absorption from the small intestine of lambs, sheep and rats. *Br. J. Nutr.* 25:57.
- 156 Windmueller, H. G., and A. E. Spaeth. 1980. Respiratory fuels and nitrogen metabolism in vivo in small intestine of fed rats. *J. Biol. Chem.* 255:107.
- 157 Wolfram, S., E. Eggenberger, and E. Scharrer. 1986. Kinetics of D-glucose and L-leucine transport into sheep and pig intestinal brush border membrane vesicles. *Comp. Biochem. Physiol.* 84A:589.
- 158 Zinn, R. A. 1990. Influence of flake density on the comparative feeding value of steam-flaked corn for feedlot cattle. *J. Anim. Sci.* 68:767.